

BACHELOR THESIS / APRIL 2024

A Multimodal Lip Recognition Approach Based on Vision and Deep Learning

Xinying Liu

Computer Science

Lancaster University College at Beijing Jiaotong University

Supervisor: Zhihai Wang

Original submission: 9 April 2024

Public portfolio copy. Administrative and signature pages omitted.

The research body, figures and references retain their original content.

Results in this thesis follow the experimental protocol described within it.

详细中文摘要

随着信息技术的迅速发展和广泛应用，个人信息识别日益成为社会关注的焦点。于是生物特征识别技术应运而生，其通过结合计算机科学、光学、声学以及生物传感器等高科技手段，利用人体固有的生理特征和行为特征进行识别。在这个背景下，唇语识别作为生物特征识别技术的重要分支之一，依托于个体的唇部形态、运动模式等特征，通过高级的模式识别和机器学习算法，实现对唇部动态的信息识别。与生物特征识别技术的其他分支相比，唇语识别技术具有非接触式、隐私性和稳定性等优势，因此在安全监控、医疗诊断、辅助通信等领域具有广泛的应用前景。

本文针对基于视觉和深度学习的多模态唇语识别方法展开研究，旨在比较不同唇语识别模型在复杂环境以及实际情况下数据集的识别准确率，调整唇语识别模型的超参数等，并提出了一种优化的耦合 3D 卷积神经网络架构。该架构能够有效地将视觉和听觉模态映射到一个统一的表示空间，并利用学习到的多模态特征来建立视听流之间的对应关系。本文具体工作如下：

首先，利用深度学习技术对唇语视频和语音等信息进行建模。通过在大规模的野生数据集上进行训练，模型能够自动学习并提取视觉和听觉模态中的关键特征。本文选取了变化度较高、野生性较强的唇语数据集 LRW-1000 以模拟实际情况的唇语识别，并进行了仿真实验对比当前热门深度学习网络模型 LSTM, LipNet, DenseNet 等在该数据集下的识别率。实验发现，多个模型在 LRW-1000 数据集上的识别率较低，并分析了复杂环境下的潜在因素。

接着，本文采用了特征级别的模型融合方法，将 3D DenseNet 模型与 LipNet 模型进行融合。两个模型的输出通过融合层进行组合，然后输入全连接层进行最终分类，以预测结果。研究采用了耦合的 3D 卷积神经网络来融合这些特征，从而实现了多模态信息的融合和交互。该网络结构不仅能够有效地捕获视觉和听觉模态之间的关联信息，还能够对口语信息进行较为准确的识别和理解。

最后，为了验证优化模型的有效性，在 LRW-1000 数据集上进行了大量多轮次的仿真实验。实验结果表明，该模型在唇语识别任务中取得了较好的收敛性能与学习效果，相较于传统方法，该模型在识别率和鲁棒性上都有了显著的提升。接着，通过对模型的优化，超参数的修改，数据集的清洗等，以优化该模型并提高模型的识别率。此外，本文还比较了不同模型在相同后端网络环境下的性能表现，所有模型与一个结构相同的后端网络相结合，该后端网络包含一个两层双向 RNN，用于最终识别。结果表明，该优化模型在复杂环境下的适应能力更强，具有更好的泛化能力。

根据大量仿真实验结果，对模型进行了深入的分析，探讨了不同参数下对识别性能

的影响，并提出了一些未来的研究方向。综上所述，本文提出的基于视觉和深度学习的多模态唇语识别方法在解决口语信息识别中的挑战方面具有重要的意义，为提高语音识别技术在复杂环境下的应用性能提供了有效的解决方案。

关键词：多模态；唇语识别；视觉识别；深度学习；卷积神经网络

Abstract

With the rapid development and wide application of information technology, personal information recognition has increasingly become the focus of social attention. As a result, biometric identification technology has emerged, which makes use of the inherent physiological and behavioral characteristics of the human body for identification by combining computer science, optics, acoustics, and biosensors and other high-tech means. In this context, lip recognition, as one of the important branches of biometric identification technology, relies on individual lip morphology, movement patterns and other features, and realises information recognition of lip dynamics through advanced pattern recognition and machine learning algorithms. Compared with other branches of biometric identification technology, lip recognition technology has the advantages of non-contact, privacy and stability, and thus has a wide range of application prospects in the fields of security monitoring, medical diagnosis, and auxiliary communication.

This paper carries out research on multimodal lip recognition methods based on vision and deep learning, aiming to compare the recognition accuracy of different lip recognition models in complex environments as well as real-world datasets, adjust the hyper-parameters of the lip recognition models, etc., and propose an optimised coupled 3D convolutional neural network architecture. The architecture can effectively map visual and auditory modalities into a unified representation space and use the learnt multimodal features to establish the correspondence between audiovisual streams. The specific work of this paper is as follows:

First, deep learning techniques are used to model information such as lip-sync videos and speech. By training on a large-scale wild dataset, the model is able to automatically learn and extract key features in both visual and auditory modalities. In this paper, the highly variable and wild lip-speaking dataset LRW-1000 is selected to simulate lip-speaking recognition in real-world situations, and simulation experiments are conducted to compare the recognition rates of the current popular deep learning network models, LSTM, LipNet, DenseNet, and so on, under this dataset. The experiments found that several models have low recognition rates on the LRW-1000 dataset and analysed the potential factors in complex environments.

Next, this paper adopts a feature-level model fusion approach to fuse the 3D DenseNet model with the LipNet model. The outputs of the two models are combined through the fusion layer and then fed into a fully connected layer for final classification to predict the results. The study employs a coupled 3D convolutional neural network to fuse these features, thus

enabling the fusion and interaction of multimodal information. The network structure is not only able to effectively capture the correlation information between visual and auditory modalities, but also able to recognise and understand spoken information more accurately.

Finally, in order to verify the effectiveness of the optimised model, a large number of multi-round simulation experiments were conducted on the LRW-1000 dataset. The experimental results show that the model achieves better convergence performance and learning effect in the lip recognition task, and compared with the traditional method, the model has a significant improvement in recognition rate and robustness. Then, the model is optimised, hyperparameters are modified, and the dataset is cleaned in order to optimise the model and improve the recognition rate of the model. In addition, this paper compares the performance of the different models in the environment of the same back-end network, where all the models are combined with an identically structured back-end network containing a two-layer bi-directional RNN for final recognition. The results show that the optimised model is more adaptable in complex environments and has better generalisation capabilities.

Based on the results of a large number of simulation experiments, an in-depth analysis of the model is carried out to explore the impact on the recognition performance under different parameters, and some future research directions are proposed. In summary, the multimodal lip recognition method based on vision and deep learning proposed in this paper is of great significance in solving the challenges in the recognition of spoken information, and provides an effective solution to improve the performance of speech recognition technology applied in complex environments.

KEYWORDS: Multi-modal; Lip recognition; Visual recognition; Deep learning; Convolutional neural networks

Contents

详细中文摘要	V
Abstract	VII
1 INTRODUCTION	1
1.1 Motivation and Background	1
1.2 Related Works	1
1.3 Objectives, Challenges and Main Contribution	3
1.4 Structure of Dissertation	5
2 COMPUTER VISION AND IMAGE GECOGNITION METHODOLOGY	7
2.1 Traditional Methods	7
2.1.1 Lip feature extraction methods	7
2.1.2 Classification and identification methods	8
2.1.3 Combining feature extraction and classification recognition	10
2.2 Deep Learning Methods	11
2.2.1 Convolutional neural network approach	12
2.2.2 Recurrent neural network method	15
2.2.3 Composite neural network method	15
3 LIP READING NETWORK	19
3.1 Datasets	19
3.1.1 Various datasets comparison	19
3.1.2 LRW-1000 selection	21
3.2 Model Architecture	22
3.2.1 LSTM mathematical model	22
3.2.2 LipNet mathematical model	24
3.2.3 DenseNet mathematical model	26
4 SIMULATION EXPERIMENTS	29
4.1 Experimental Design	29
4.1.1 Data preprocessing	29
4.1.2 Implementation of lip sequence conversion algorithm	32
4.1.3 Experimental methods	33
4.2 Results Comparison	34
4.2.1 LSTM model	34
4.2.2 DenseNet model	35

4.2.3 Model comparison	39
4.3 Optimization Approaches	40
4.3.1 Model tuning hyperparameter optimization	40
4.3.2 Model optimization strategy	42
4.3.3 Optimised model training rate	46
5 CONCLUSIONS	47
5.1 Review of Aims	47
5.2 Future Work	48
References	49
Acknowledgements	52
Appendixes	65

1 INTRODUCTION

The introduction section will outline the research background and motivation of the project and summarize the current state of research based on the literature review. Then, the applications and challenges of current technology research will be presented. In addition, this chapter will outline the overall structure of the thesis.

1.1 Motivation and Background

With the development and progress of society, speech recognition technology is becoming more and more important in people's lives. From various smart platforms to smart homes, speech recognition technology has become one of the most vital ways in daily human-computer interaction. However, in real life, speech recognition systems often face a variety of display challenges, such as environmental noise, inarticulation, speaking habits, etc., which largely reduces the accuracy of speech recognition and thus affects its scope of application. To improve the robustness and accuracy of speech recognition, researchers have started to explore multimodal speech recognition methods, and one of the directions is visual lip recognition.

In today's digital society, the rapid development of AI technology and deep learning methods makes lip recognition technology gradually become a research hotspot. Lip recognition is a technology that integrates machine vision and natural language processing. Through vision processing to extract frames continuously from the video, and then from multiple frames of the image to identify the face separately, through the image changes to derive the corresponding out of the continuous mouth changes, audio information, word information, etc., the information will be input into the lip recognition model for learning, so as to identify the pronunciation corresponding to the mouth of the face video.

1.2 Related Works

Lip reading (also called VSR) is the process of converting a video or image sequence of the lips during human speech into a sequence of words. In 1976, the McGurk effect revealed that the brain produces a third perceptual phoneme when visual and auditory information do not match up. This suggests that in verbal communication, the brain processes both visual and auditory information, combining the two to obtain more comprehensive linguistic information

[24]. Although speech recognition technology has been more mature, to further improve the ability of computer recognition of language, it is necessary to conduct more in-depth research in the field of lip recognition. Lip recognition not only effectively assists speech recognition, but also serves as an independent language recognition method when speech information is difficult to obtain or when noise interference is high [4]. The following chart Figure 1 shows the evolution of lip recognition, outlining the process from its infancy all the way to today's popular recognition methods [18][31].

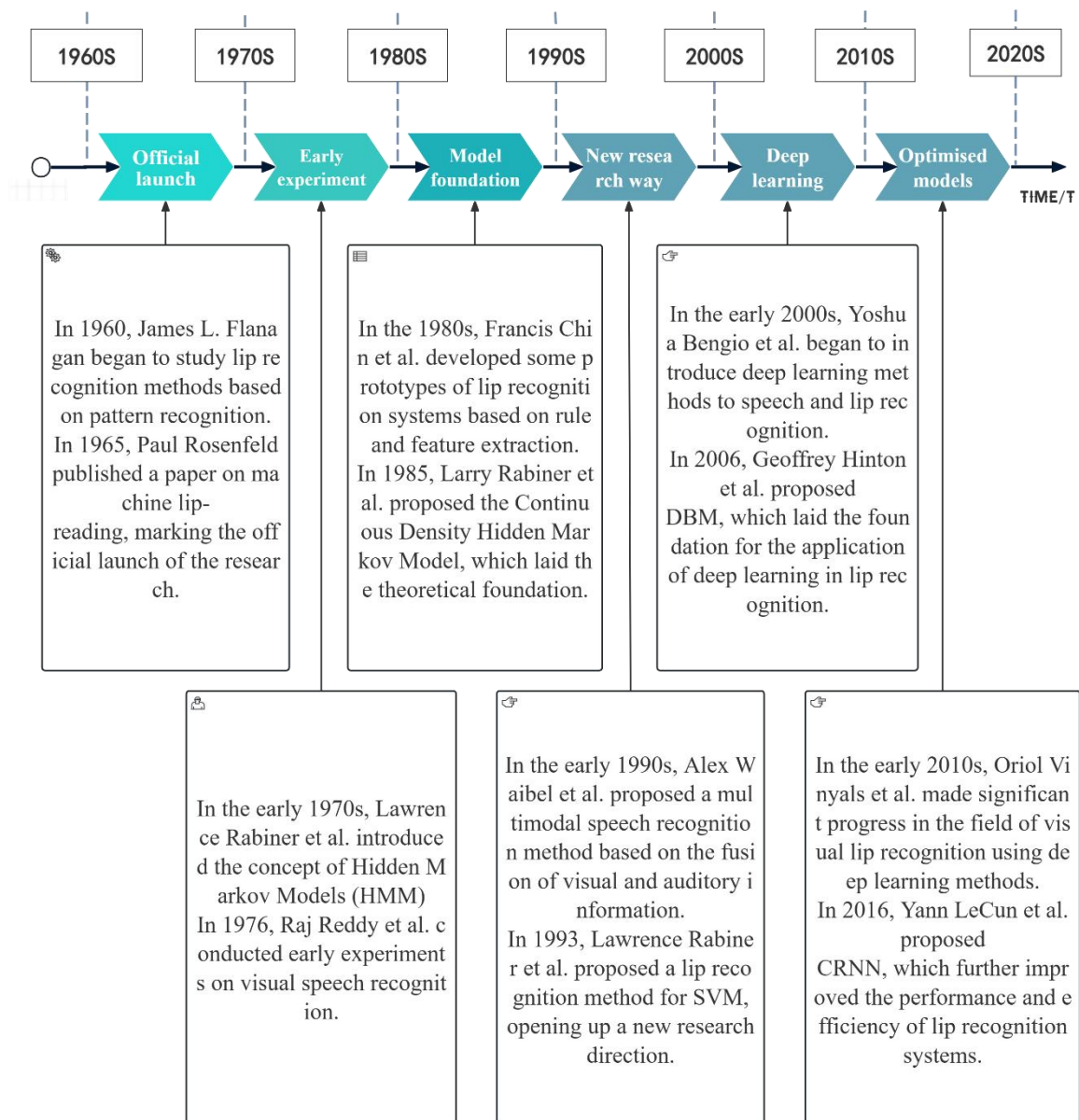


Figure 1 Lip reading milestones of development

In recent years, many researchers have explored lip recognition using deep learning methods. In 2016, Assael et al. from the University of Oxford proposed the first end-to-end sentence-level lip recognition method, which uses a 3D spatio-temporal convolutional network STCNN to extract features, a bi-directional GRU network to classify the phonemes, and a combination of a CTC loss function to combine the phonemes into complete sentences. The method achieved a word error rate (WER) of 4.8% on the GRID dataset [2]. In the same year, Chung et al. presented the first fully automated method for capturing lip-speaking data from videos of natural environments and compared four CNN-based lip-recognition methods, the best of which achieved 91.4% accuracy on the OuluVS1 dataset and 93.2% on the OuluVS2 dataset [8]. Represented by these two papers, many papers have taken a similar approach to develop various deep learning lip recognition methods based on CNNs and LSTMs [1] [28] [29] [30].

Currently, most lip recognition-based methods use CNNs to extract lip features. Although neural networks are capable of automatically extracting abstract features suitable for classification, morphological changes in lip features are critical in lip recognition. Therefore, extracting features directly using CNNs is susceptible to the influence of other irrelevant features [33]. In addition, the pooling operation in CNNs is not applicable to extract the edge information of an image, which further increases the challenge of lip recognition [3].

1.3 Objectives, Challenges and Main Contribution

Acquiring and processing multi-modal information is becoming increasingly convenient. Lip reading has emerged as a widely used communication method for understanding a speaker's pronunciation and speech content by observing their lip movements. Lip recognition technology, as part of multi-modal information fusion, has demonstrated its significance across various fields. Today, multi-modal lip recognition technology holds vast potential for application. For instance, in the realm of security authentication, multi-modal lip recognition offers high-security biometric authentication. It finds application in employee attendance systems and access control systems. By authenticating employees through lip movement recognition, contactless attendance and access control management are achieved, circumventing the security risks and hygiene concerns associated with traditional card or fingerprint recognition systems. Additionally, in multimedia content generation, technology based on multi-modal lip recognition can generate subtitles for audio and video content, enabling access to a broader range of languages [19]. Moreover, in entertainment, the

technology can enhance Virtual Reality (VR) and Augmented Reality (AR) gaming experiences by recognizing players' lip movements, facilitating a more immersive and interactive gaming experience ^[17]. In industrial applications, lip recognition technology can enhance the efficiency of production lines by identifying and rectifying errors through the monitoring of workers' lip movements.

However, several significant challenges accompany the application of multimodal lip recognition technology. Firstly, small variations in lip movements may affect recognition accuracy, necessitating improvements in sensitivity and accuracy to lip movements. Secondly, environmental factors such as lighting conditions and background noise may lead to recognition errors. Furthermore, cross-language and accent differences pose a challenge, highlighting the need to enhance the system's capability to recognize diverse languages and accents. Lastly, acquiring, and labeling datasets present a major challenge, necessitating the creation of large-scale, representative lip-reading datasets to ensure adaptability and robustness.

Therefore, in this paper, research on multimodal lip recognition methods based on vision and deep learning is carried out with the aim of comparing the recognition accuracies of different lip recognition models in complex environments as well as real-world datasets, adjusting the hyperparameters of the lip recognition models, etc., and proposing an optimised coupled 3D convolutional neural network architecture. The research in this paper includes:

(1) Comparison of various aspects of the nature of multiple lip recognition datasets, and selection of a wild dataset that best fits the reality and actual environment for subsequent experiments.

(2) The conduction of simulation experiments to compare the recognition rates of current popular deep learning network models LSTM, LipNet, DenseNet, etc. under this dataset. and the analyzation of the potential factors of the species in the complex environment.

(3) The design of A novel coupled 3D convolutional neural network architecture to map visual and auditory patterns into a unified representation space.

(4) Data cleaning and augmentation, etc., and various hyper-parameters of the model adjustment to optimize the model to improve the lip recognition rate. The learned multimodal features are used to establish the correspondence between the visual and auditory streams to achieve the goal of visual lip-sync recognition rate.

(5) Conduction of simulation experiments on the LRW-1000 dataset to validate the performance of the model proposed in this paper and comparison and analyzation of it with existing methods. and give the subsequent model optimization scheme.

This research is carried out with the expectation of proposing an effective visual lip-synthesis recognition method, which will provide new ideas for the application of speech recognition systems in complex environments. This will help to improve the robustness and accuracy of the speech recognition system and promote the further development of speech recognition technology in practical applications.

1.4 Structure of Dissertation

The aim of this thesis is to explore multimodal lip recognition methods based on vision and deep learning to achieve the goal of accurate recognition and understanding of spoken messages in complex environments. I would like to propose a coupled 3D convolutional neural network architecture that can map two patterns to a representation space and use the learned multimodal features to determine the correspondence between audiovisual streams.

The main structure of this paper is as follows:

(1) The first part firstly introduces the motivation of the research and describes the use of lip recognition and the current challenges encountered, pointing out the importance of lip recognition in practical applications.

(2) The second part explores the development history of lip recognition and the related work done by previous authors and reviews the current research status and progress in multimodal data fusion, lip image feature extraction, and deep learning model design.

(3) The third part introduces the implementation method and related theoretical foundation of this paper and analyses the specific implementation process of traditional methods and deep learning methods in detail.

(4) The fourth part describes in detail the dataset and lip recognition network structure used and designed in this paper, conducts a comparative analysis, and proposes an analysis of the relevant models as well as their advantages and disadvantages, and proposes an optimization strategy based on the comparison.

(5) The fifth section shows a series of experiments on the selected network model, analyses the experimental results, and optimizes the data results of different deep learning

methods accordingly.

Finally, the conclusion section summarizes the experimental results of this paper and present the outlook for further research on lip recognition in the future.

2 COMPUTER VISION AND IMAGE GECOGNITION METHODOLOGY

Traditional lip recognition methods have a long history of development, but due to the complexity of the lip recognition problem and the limitations of traditional methods, it is difficult to achieve truly practical research results. In recent years, with the rapid development of deep learning technology, researchers have made great progress by applying deep learning methods to the field of lip recognition. This section presents all the theoretical foundations used in the research. Firstly, several traditional lip recognition methods are briefly introduced, and then the deep learning-based lip recognition methods are elaborated in detail.

2.1 Traditional Methods

Traditional lip recognition methods can be divided into three steps: lip detection and localization, lip feature extraction and classification recognition. This type of algorithm has a long history of development, with many related research results and corresponding reviews, so this paper does not systematically introduce this type of methods, but only briefly introduces several algorithms that have good recognition results on common datasets in recent years.

2.1.1 Lip feature extraction methods

In 2009, Papandreou et al of the IEEE proposed a multi-modal fusion feature extraction scheme capable of dynamically adapting to changes in environmental conditions and applied it experimentally to the problem of speech recognition with audio-video fusion. This multi-modal fusion scheme combines a six-weighting approach with an adaptive compensation mechanism, where stream weighting provides a bias towards the most informative modes and uncertainty compensation provides a fine-grained adaptation mechanism that considers different environmental conditions. For video recognition, a speaker adaptation mechanism is used that eliminates the inherent appearance variations of neutral pose multi-face images that are not related to visual speech^[26].

The dataset used for the experimental part is the CUAVE dataset from Clemson University (2002). The face of the speaker's is first tracked, and visual features are extracted from it using AAMs, which capture the size and quality variations of the face in contrast to

image transformation-based methods such as DCT/PCA/DWT/ICA that process pixels of facial images. The special feature here is the use of a cascade of two AAMs. The first full-face AAMs allows accurate tracking of the face region in long video sequences, and then a second ROI-AAM is used to cover the mouth region and analyse the shape and texture of the ROI. Afterwards, the shape-texture averages for a specific person are subtracted from the feature values to avoid recognition that focuses too much on the identity of the speaker and ignores the linguistic information in the video [27].

In the experimental part of video recognition, three AAM methods with different parameters are compared with PCA method and the experimental results are shown in Figure 2. From the experimental results, the AAM method with six shape and six texture coefficients has the best results. This experimental result is the highest recognition rate on the CUAVE dataset.

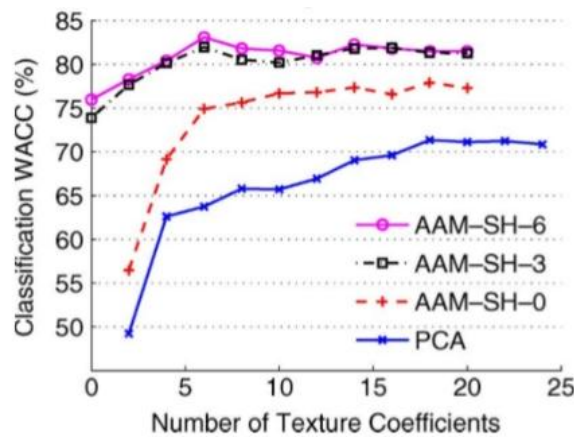


Figure 2 Comparison of classification results of digital lip recognition by AAM and PCA methods with different texture coefficients

2.1.2 Classification and identification methods

In 2016, Hu et al from Northwestern Polytechnical University proposed a Recurrent Temporal Multimodal Restricted Boltzmann Machine (RTMRBM) as a classification method for lip recognition [15]. The RTMRBM algorithm has the following three advantages:

- (1) The algorithm can extract semantic information from each modal data and learn a joint representation across audition modalities.
- (2) It is a directed graph model that models temporal audition sequences well because the joint representations between all frames are correlated.

(3) Simple connections between frame sequences and standard MRBM are easy to train.

Three datasets are used in the experimental part, namely the publicly available datasets AVLetters and AVLetters2^[9], and the home-made dataset AVDigits. the latter two datasets are only experimented with combined audio-video lip-recognition, whereas the present review addresses video-only lip-recognition methods, so only the datasets are listed here Experimental results for AVLetters. In processing the video image, the cascade object detector is first used to extract the key region of the lips, and this part of the image is scaled to 60*80 pixels, and then reduced to 100 principal components using the PCA method. Then the RTMRBM algorithm is used for training.

Experimental results on the AVLetters dataset show that the RTMRBM algorithm exhibits significant advantages when comparing with existing recognition methods based on manual features or multimodal deep networks. The experimental results are shown in Table 1. It can be seen that the RTMRBM algorithm achieves the highest recognition accuracy when only audio information, video information or both audio and video information are utilised. This experimental result highlights that the RTMRBM algorithm can achieve the best recognition performance on the AVLetters dataset, is highly adaptable in cross-modal lip recognition tasks and is able to effectively exploit the complementary nature of audio and video information.

Table 1 Mean Accuracy of RTMRBM and Other Methods for Speech Recognition in Three Modes on AVLetters Dataset

Modality	Model	Mean Accuracy
A	MDAE	58.40
	CRBM	61.2
	RTMRBM	64.41
V	Multiscale Spatial Analysis	44.6
	Local Binary Pattern	58.85
	3DCNN-HMM	59.6
	MDAE	62.10
	CRBM	62.6
	RTMRBM	64.63
AV	MDAE	62.9
	CRBM	64.8
	RTMRBM	66.4

2.1.3 Combining feature extraction and classification recognition

In 2008, Fu et al. of IEEE proposed a subspace learning algorithm called discriminant simplex analysis (DSA, discriminant simplex analysis) for feature extraction; and a classification criterion based on the concept of nearest neighbors (kNNS) for classification. The authors applied this algorithm to facial recognition and lip recognition and conducted experiments^[11]. The DSA algorithm has the following four advantages:

- (1) The algorithm learns the graphical parameters automatically without any arbitrary settings.
- (2) Only two parameters are fine-tuned in the solution.
- (3) The algorithm minimizes the distance between each sample and the nearest point in the same category, while maximizing the distance from each sample to the nearest point in a different category.
- (4) The distance between each sample and a sample in a category provides a more accurate and reliable measure of its likelihood of belonging to a category.



Figure 3 Comparison of the highest lip-reading refinement and downscaling effects of the seven methods for each HMM state under four different amounts of Gaussian mixing

The dataset used in the experimental part is the AVICAR dataset^[20]. The Adaboost-based face tracker and lip tracker were first used to circle the lip region, and then the full image of

this region was used to create the source high-dimensional feature space. A DCT method is used to compress the feature vector to 100 dimensions, and then seven different subspace learning methods (DCT, PCA, LDA, LPP, LEA, LSDA, DSA) are used to represent the reduced 1–50-dimensional lip-reading features. Finally, a left-to-right continuous density Hidden Markov Model (HMM) is trained on the word level in the training database.

Table 2 Comparison of the highest lip-reading refinement and downscaling effects of the seven methods for each HMM state under four different amounts of Gaussian mixing

	2	4	8	32
DCT	35.87	35.9	36.76	36.73
PCA	35.8	36.63	36.91	36.73
LDA	36.02	36.91	36.48	36.6
LPP	34.61	36.73	35.72	35.57
LEA	36.38	36.65	37.08	37.21
LSDA	36.38	36.86	36.81	36.76
DSA	36.91	37.41	37.87	37.76

The experimental results are shown in Figure 3 and Table 2. From the experimental results, it can be observed that the graph-based stream learning dimensionality reduction methods (LEA, LSDA, DSA) outperform the (DCT, PCA, LDA) linear subspace learning algorithms for different number of Gaussian mixtures. The dimensionality reduced DSA algorithm has the highest lip-reading accuracy in all four cases and requires a similar number of dimensions. This experiment results in the highest recognition rate on the AVLetters dataset.

2.2 Deep Learning Methods

Deep learning has been widely used in many fields, among which the more commonly used neural networks include convolutional neural network and recurrent neural network, among which convolutional neural network (CNN) is mostly used for processing images, extracting deep features in the image through convolutional operations on individual pixel points in the image, to carry out the recognition and classification. Recurrent neural network (RNN for short) is mostly used to deal with continuous sequences, by calculating the information of the previous nodes in the sequence and applying it to the calculation of the current node, it is able to extract the relationship between the continuous sequences as the features and carry out the classification and recognition work.

Through the above introduction, it can be seen that the processing of lip-speaking video is related to both network structures, CNN is able to extract the features in each frame of the lip-speaking image, and RNN is able to extract the relationship between each frame of the lip-speaking image, so the researchers have carried out experimental studies on both network structures during the research of lip-speaking recognition methods, and also combined the two kinds of neural networks, to take full advantage of the functions of the two kinds of network structures. The researchers have also combined the two neural networks to make full use of the functions of the two network structures, and to achieve a more practical lip recognition method. In the following, the method of using two kinds of neural networks and the method of combining two kinds of neural networks to propose a composite neural network are introduced respectively.

2.2.1 Convolutional neural network approach

Convolutional neural networks are commonly used to process images, where a single image is a two-dimensional (2D) piece of information. A basic 2D convolutional layer from channel C to channel (without bias and unit step) is calculated as follows:

$$[conv(x, w)]_{c'ij} = \sum_{c=1}^C \sum_{i'=1}^{k_w} \sum_{j'=1}^{k_h} w_{c'ci'j'} x_{c,i+i',j+j'} \quad (1)$$

Where for input x and weights $w \in \mathbb{R}^{C' \times C \times k_w \times k_h}$, where denotes i, j exceeds the bounds.

In lip recognition work, since the information being processed is three-dimensional (2D) lip video, some researchers have also used spatio-temporal convolution (3D convolution) approach, spatio-temporal Convolutional Neural Networks (STCNNs) can be used to process the video data by convolving across time as well as spatial dimensions. Their calculation formula is as follows.

$$[stconv(x, w)]_{c'tij} = \sum_{c=1}^C \sum_{t'=1}^{k_t} \sum_{i'=1}^{k_w} \sum_{j'=1}^{k_h} w_{c'ct'i'j'} x_{c,t+t',i+i',j+j'} \quad (2)$$

The first application of CNN to the field of lip recognition was proposed by Noda et al in Japan in 2014. This paper used a CNN network containing 3 convolutional layers, 3 pooling layers and one fully connected layer to convert a lip image of size 32*32-pixel points into a 40-dimensional vector as the extracted lip features. The features are classified and recognized by an HMM-GMM model for independent lip word recognition. The experimental dataset is a

homemade Japanese dataset containing 300 Japanese words. The experimental results show that the CNN feature extraction method has better experimental results compared to the PCA method, and the simple pixel point compression method [24].

In 2016, Saitoh et al. from Japan and Finland proposed a CNN-based lip recognition method. In order to make full use of the CNN's ability to extract image features, so that the CNN network can extract the features of lip movement transformations in lip-speaking videos in a single image, this paper proposed a cascaded frame image (CFI) method, which is centered on splicing multiple frames of lip images into a single image in a chronological order, so that one image covers the whole process of lip movement, and then three different CNN networks, NIN, AlexNet and GoogLeNet, are used for feature extraction and classification [28]. The method flow is shown in Figure 4. The experiments were carried out using the OuluVS2 dataset on multiple viewpoints separately, and high recognition accuracy was achieved.

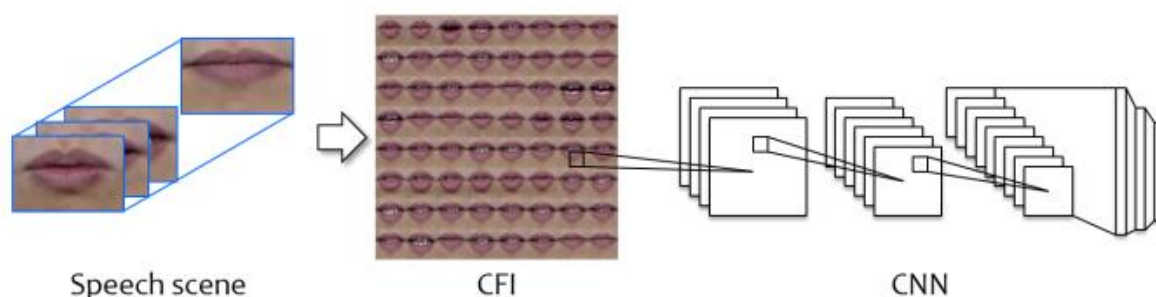


Figure 4 Overview of Proposed framework

In 2016, Chung et al from the University of Oxford presented proposed and compared four different CNN network architectures and compared their advantages and disadvantages experimentally. There are two basic divisions between the four architectures: the timing of the multi-frame image fusion and the dimensionality of the convolution kernel. The EF-3(3D Convolution with Early Fusion) and EF (Early Fusion) architectures fuse multi-frame images at the beginning of the network; this early fusion architecture is a common approach used in similar CNN architectures before, which requires accurate connections between frames, and there is no strict alignment between frames when intercepting the lip region in the lip-speaking experiment [6] [8]. On the contrary, the MT-3 (3D Convolution with Multiple Towers) architecture and the MT (Multiple Towers) architecture first performs a set of convolutions and pooling operations on each image separately before fusion, which reduces the impact of differences between image frames on the network. The disadvantage is that the same filter weights are used for all frames in the common transformation layer, whereas the

EF architecture has different weights for each frame. 2D convolution is used in both EF and MT architectures, while 3D convolution is used in EF-3 and MT-3 architectures. The specific structures of the four architectures are shown in Figure 5 [7]. Theoretically, the 3D convolution should be better considering the time factor, but the experimental results show that 2D convolution is better than 3D convolution. results are better than 3D convolution. The experimental datasets used are the publicly available datasets OuluVS1, OuluVS2 and the home-made LRW dataset. The concrete convolutional neural network framework and architecture is shown in Figure 5 below. From the experimental results, the MT architecture has the highest recognition accuracy. The main classification errors occur between words and their plural forms, or between words that cannot be distinguished by virtue of visual information [4].

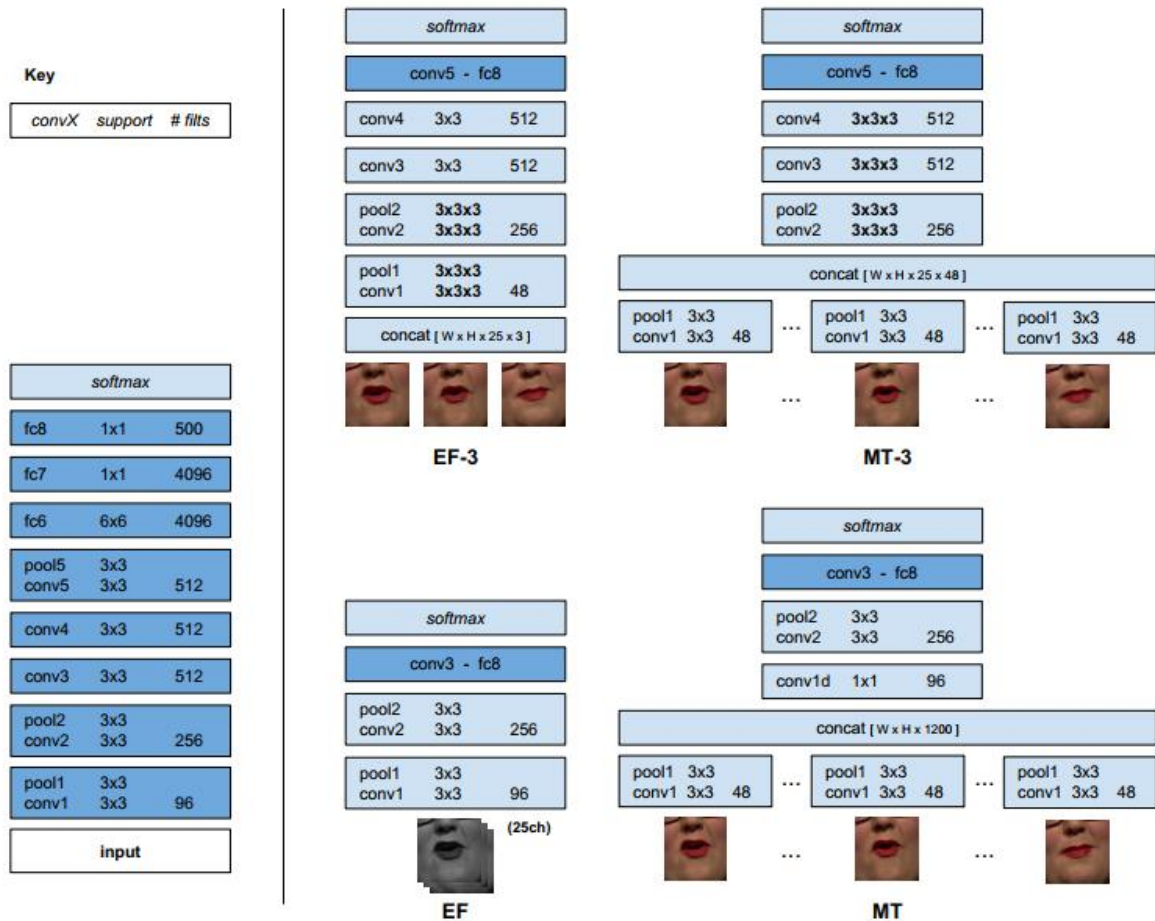


Figure 5 Convolutional Neural Network Architecture

2.2.2 Recurrent neural network method

Recurrent neural networks are mostly used to process continuous sequences. However, the structure of the initial RNN model is complex, and the improved versions of RNN networks are usually used: Long Short-Term Memory (LSTM) ^[14] or Gated Recurrent Unit (GRU). The mathematical model of RNN networks is introduced here using GRU as an example:

$$\begin{aligned} [u_t, r_t]^T &= \text{sigm}(W_z z_t + W_h h_{t-1} + b_g) \\ \tilde{h}_t &= \tanh(U_z z_t + U_h (r_t \odot h_{t-1}) + b_h) \\ h_t &= (1 - u_t) \odot h_{t-1} + u_t \odot \tilde{h}_t \end{aligned} \quad (3)$$

where $z := \{z_1, \dots, z_T\}$ is the input sequence of the RNN, $\odot$ denoting the elemental multiplication, $\text{sigm}(r) = 1 / (1 + \exp(-r))$. Bidirectional GRU (Bi-GRU), introduced in the context of LSTM (1997), is commonly used in lip recognition models: one RNN mapping is $\{z_1, \dots, z_T\} \mapsto \{\vec{h}_1, \dots, \vec{h}_T\}$, the other is $\{z_T, \dots, z_1\} \mapsto \{\bar{h}_1, \dots, \bar{h}_T\}$, and $h_t := [\vec{h}_t, \bar{h}_t]$. the Bi-GRU ensures that for all t , h_t that depends on the $z_{t'}$. To parameterize the distribution over the sequence, at time step t , such that $p(u_t | z) = \text{softmax}(\text{mlp}(h_t; W_{mlp}))$, where NLP is a feedforward neural network with weights W_{mlp} . We then define the distribution over sequences of length T as, where T is determined by the input z to the GRU.

In 2016, Wand et al from Switzerland proposed an LSTM-based lip recognition method. This method uses two LSTM layers with 128 units per layer to directly classify and recognize a 40*40 sequence of lip images. The experiment was performed on 51 words or letters from the GRID dataset and achieved the highest recognition accuracy at the word level on the GRID dataset ^[32]. The experimenter has also conducted experiments on LSTM model with the addition of CNN network, but the experimental results showed that adding CNN network did not improve the performance, the reason is that the lip image of 40*40-pixel points is already small enough, and it is difficult to extract more effective information using CNN.

2.2.3 Composite neural network method

Both CNN and RNN networks have their own advantages, combining the two, using CNN to extract the features of a single-frame lip image and RNN to extract the relationships between the lip image sequences, the resulting composite neural network can extract the features in a better way and improve the performance of the lip recognition method.

In 2016, Assael et al from the University of Oxford proposed a sentence-level lip

recognition method [2]. Previous lip recognition methods are limited by classification methods that can only classify consecutive lip sequences into a single category, so they can only achieve lip recognition for types such as letters, words, phrases, and so on. In order to make lip recognition extend from words to longer sentences, this thesis draws on the connectionist temporal classification (CTC) loss method commonly used in modern speech recognition techniques [13]. For a given model, which outputs a series of discrete distributions over a class of tokens (a vocabulary) augmented with special 'blank' markers, CTC calculates the probability of a sequence by marginalizing all sequences that are identical to it. This approach allows longer sequences to be recognized without alignment and the length of the sequence can be varied. Let V denote the set of values taken by the model classification at a single time step of its output (the word series), which is then the blank augmented vocabulary $\tilde{V} = V \cup \{\}$, where $\{\}$ denotes the CTC blank symbol. Define a function that $B : \tilde{V}^* \rightarrow V^*$, for an exceeded string $\tilde{V}$, removes neighboring duplicate characters and removes blank tags. For a sequence of tags $y \in V^*$, CTC defines.

$$P(y | x) = \sum_{u \in B^{-1}(y) \text{ s.t. } |u|=T} P(u_1, \dots, u_T | x) \quad (4)$$

where T represents the number of time steps in the sequence model. For example, if $T = 3$, CTC defines the probability of the string "am" as $ab \rightarrow P(aab) + P(abb) + P(ab) + P(a b) + P(ab)$. This aggregate is excuted by dynamic programming for its maximum likelihood. This paper combines spatio-temporal convolution, bi-directional GRU and CTC loss, the structure of which is illustrated in Figure 6. Experiments were conducted on the GRID dataset, which achieved the best recognition results on the GRID dataset at the time.

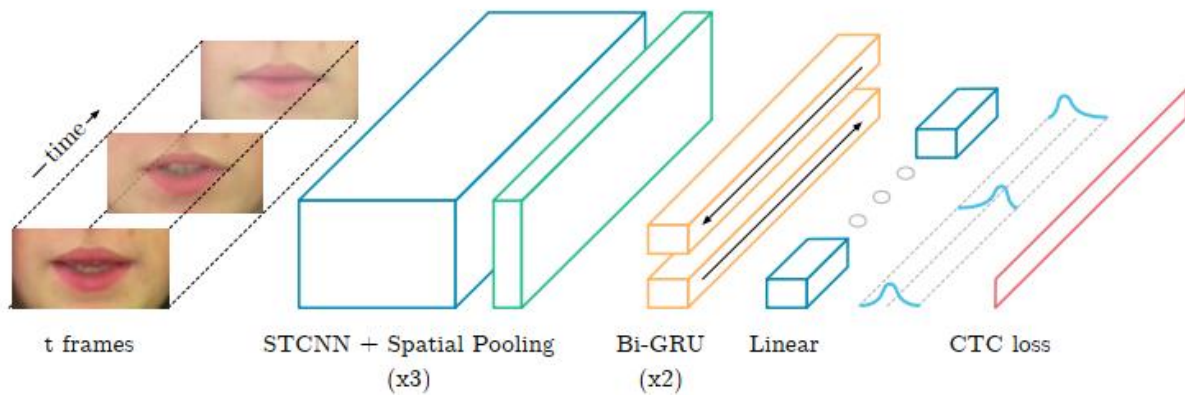


Figure 6 Composite neural network processing overview

In 2017, Chung et al from the University of Oxford proposed a combined audio-video lip recognition method. To solve the sentence-level lip recognition problem, this article used an encoder-decoder (encoder-decoder) model, the input sequence is read, followed by the prediction of the output sequence.. This model structure is divided into 3 parts, video feature extraction (encoding), audio feature extraction (encoding) and classification recognition (decoding). The video encoding part uses a combined CNN and LSTM network to obtain lip video feature sequences, the audio encoding part directly uses an LSTM network to obtain audio feature sequences, and the decoding part uses an LSTM network that combines a dual-attention mechanism, in which the video information and the audio information are controlled by two independent attention mechanisms, which can solve the problem of the lack of temporal alignment of the two modalities and also make the two modalities more efficient [5]. This can solve the problem that the two modalities are not temporally aligned, and also make the two modalities independent of each other, so that they can be recognised in the absence of one of the input streams, i.e., only audio or only video. The specific structure of the model is shown in Figure 7. The paper conducted experiments on the self-made LRS dataset and GRID dataset and achieved good experimental results. In addition, the authors also experimented with bidirectional LSTM on the video coding and audio coding parts of the network model, which increased the training time but did not improve the performance, probably because the attention mechanism of the decoder module is already sufficient to extract the key features, and bidirectional LSTM is not necessary.

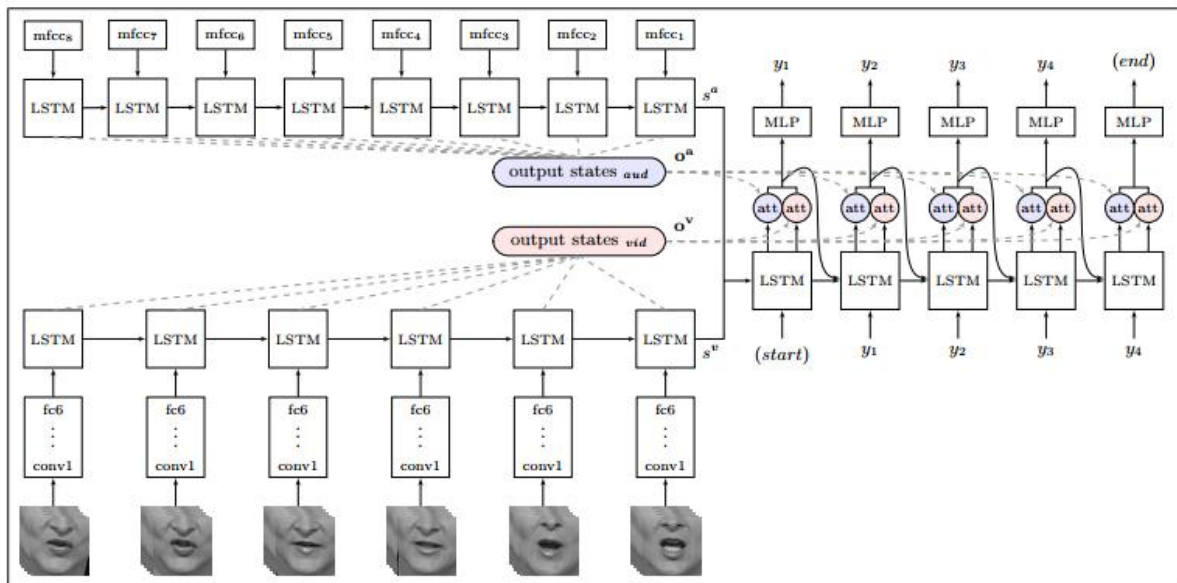


Figure 7 Composite neural network processing frame

In 2018, Afouras et al. from the University of Oxford compared several lip recognition models and investigated online (real-time) lip recognition models that can be used for continuous lip videos. The three models proposed by the authors are Bidirectional LSTM (BL) model, Fully Convolutional (FC) model and Transformer model (TM). The BL model combines LSTM and CTC but does not include the CNN network part; the FC model combines CNN and CTC, including a model FC-10 with 10 convolutional layers and a model FC-15 with 15 convolutional layers. The TM model adopts an encoder-decoder structure, which adopts the encoder-decoder structure similar to that in the paper ‘Attention is all you need’^[31] in the form of superimposed 6-layer encoder and 6-layer decoder with self-attention mechanism. The specific structures of the three models are shown in Figure 8. Experimental results on the LRW and LRS2 datasets show that the FC model has a smaller number of parameters and faster training speed, the BL model has a poorer performance, and the TM and FC-15 models have the same time complexity for each layer, but the TM model requires a lot of iterations, and the total training time is longer^[1]. Since the FC model uses spatio-temporal convolutional operations, it is possible to artificially control the length of the context that the model can see. On the contrary, when the LSTM model is used, the complete input sequence needs to be acquired before recognition classification. Using this feature of the FC model, it can be used for low-latency online lip recognition.

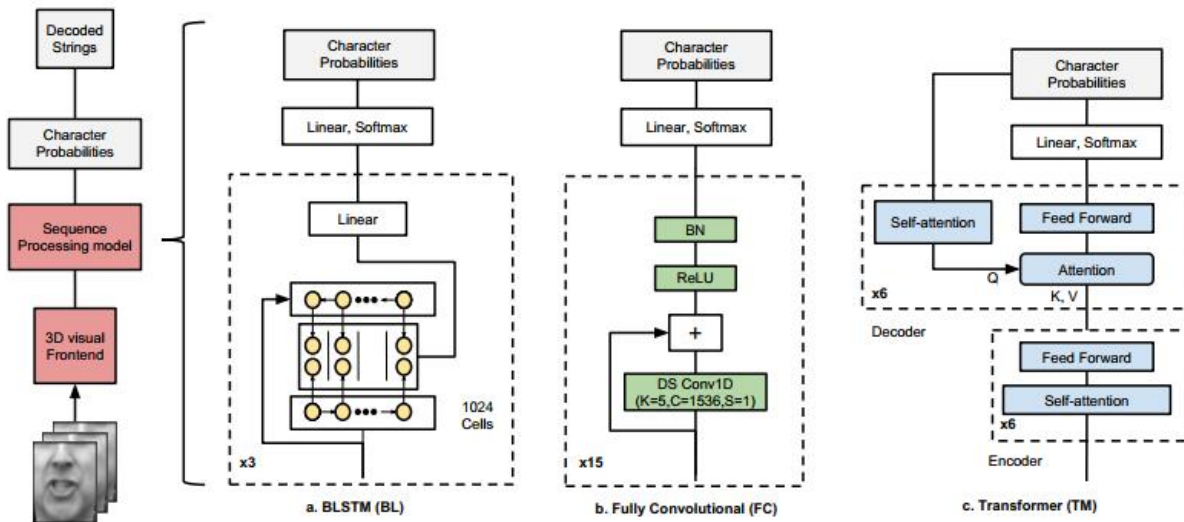


Figure 8 Lip reading model, the image service is then processed by one of three architectures. (a) BL: recursion of the bridge of a set of depth-separated tensors for the multi-terminal environment transformation model LSTM layers; (b) FC: full key, value and key (c) TM

3 LIP READING NETWORK

This chapter focuses on the dataset selection process and the lip recognition model employed. Specifically, the performance of several publicly available datasets is first compared in various aspects to identify the most suitable field dataset for the real-world context; subsequently, the mathematical model of the adopted network model and its implementation are described in detail.

3.1 Datasets

Lip-speaking datasets are an important foundation for the study of lip-recognition techniques, and the collection of lip-speaking data is difficult, and the number of publicly available lip-speaking datasets is relatively small, and the collection and organisation of lip-speaking datasets is also an important part of lip-recognition research. This section describes the selection of lip-speaking datasets.

3.1.1 Various datasets comparison

This subsection compares the information of some existing lip-speaking datasets. Several commonly used lip-speaking datasets are introduced below.

The AVICAR dataset records video and speech information in an automotive environment, capturing information from eight microphones on the sun visor and four cameras on the dashboard. In order to vary the signal-to-noise ratio, each utterance was subjected to five different noise environments. The CUAVE dataset consists of two parts: a single speaker and a pair of speakers. The speakers were selected so that there was a balanced ratio of men to women and there were different skin colors and accents, as well as visual features such as glasses, beards, hats, etc. The OuluVS1 dataset consists of 20 speakers each reading ten phrases aloud, each phrase being read five times. The speakers included 17 males and 3 females from four countries. The OuluVS2 dataset is an upgraded version of the OuluVS dataset. This dataset includes 53 speakers, including 40 males and 13 females, all of whom are non-native speakers of English and come from regions such as Europe, China, India, Arabia, and Africa. Six cameras were used to capture the video information simultaneously. The GRID dataset consists of 34 speakers, each speaking 1000 sentences, for a total of 34,000 sentences over 28 hours. The sentences in this dataset have a specific

grammar, i.e. a command + color + preposition + letter + number + adverb. The native language of all is English. The video source for the LRW dataset is BBC News, which consists of regular news with a single presenter facing the camera, and panel debates with multiple speakers exchanging realizations. The resulting LRW dataset consists of 500 of the most frequently occurring words with a length of 5-10 letters, which is chosen so that the video length is less than 1 second, while words that are too short are more ambiguous and more difficult to judge visually. The LRW-1000 dataset^[34] is an English-annotated dataset of Chinese lip-syncing. The data source for the LRW-1000 dataset is a set of 51 television programs from 26 TV stations, with video genres including broadcast news and dialogue programs, and more than 2,000 speakers, including program hosts, guests, reporters, and interviewees of different genders and ages who speak Mandarin or various dialects. The valid lip-sync videos included 745,187 samples in 1,000 phrase categories, with each phrase including an average of 750 samples. The average length of each video was 0.3 seconds. The concrete datasets Identifications, sizes, environments and their recognition rate are shown in Table 3.

Based on the above data comparison and analysis and combined with the actual situation of the research in this paper, the dataset of this paper uses LRW-1000. The reasons for choosing this dataset are explained in detail in the next subsection.

Table 3 Comparison of lipreading datasets

Dataset Name	Acquisition environment	Identification of content	video angle	Data size	Highest recognition rate
AVICAR	In-car	Numbers	Positive	851	37.9%
AVLetter	In-laboratory	Letters	Positive	78	65.13%
CUAVE	In-laboratory	Numbers	Positive	1800	83.0%
OuluVS1	In-laboratory	Phrases	Positive	200	91.4%
OuluVS2	In-laboratory	Phrases	Multi-angle	520	95.6%
GRID	In-laboratory	Sentences	Positive	28775	95.2%
LRW	TV program	Words	Positive	400000	83.0%
LRS2	BBC	Sentences	Positive	118116	66.5%
LRS3	TED	Sentences	Positive	165000	62.8%
LRW-1000	TV program	Words	Multifaceted	745187	38.19%

3.1.2 LRW-1000 selection

The LRW-1000 dataset (2019) is a 1,000-level Chinese lip-sync dataset provided by the Institute of Computing Technology, Chinese Academy of Sciences (ICTC). The data sources for the LRW-1000 dataset are 51 TV programs from 26 TV stations. The video genres include broadcast news and dialogue programs. Lip-speaking videos were collected from more than 2,000 speakers, including program hosts, guests, reporters and interviewees of different genders and ages who speak Mandarin or various dialects. The valid video set contains a total of 718,018 samples, covering more than 1 million instances of Chinese characters and 286 Chinese syllables. Each class corresponds to a syllable of a 10,000-character word, where a word consists of one or more Chinese characters. Some of the examples in the LRW-1000 are shown in Figure 9 below.

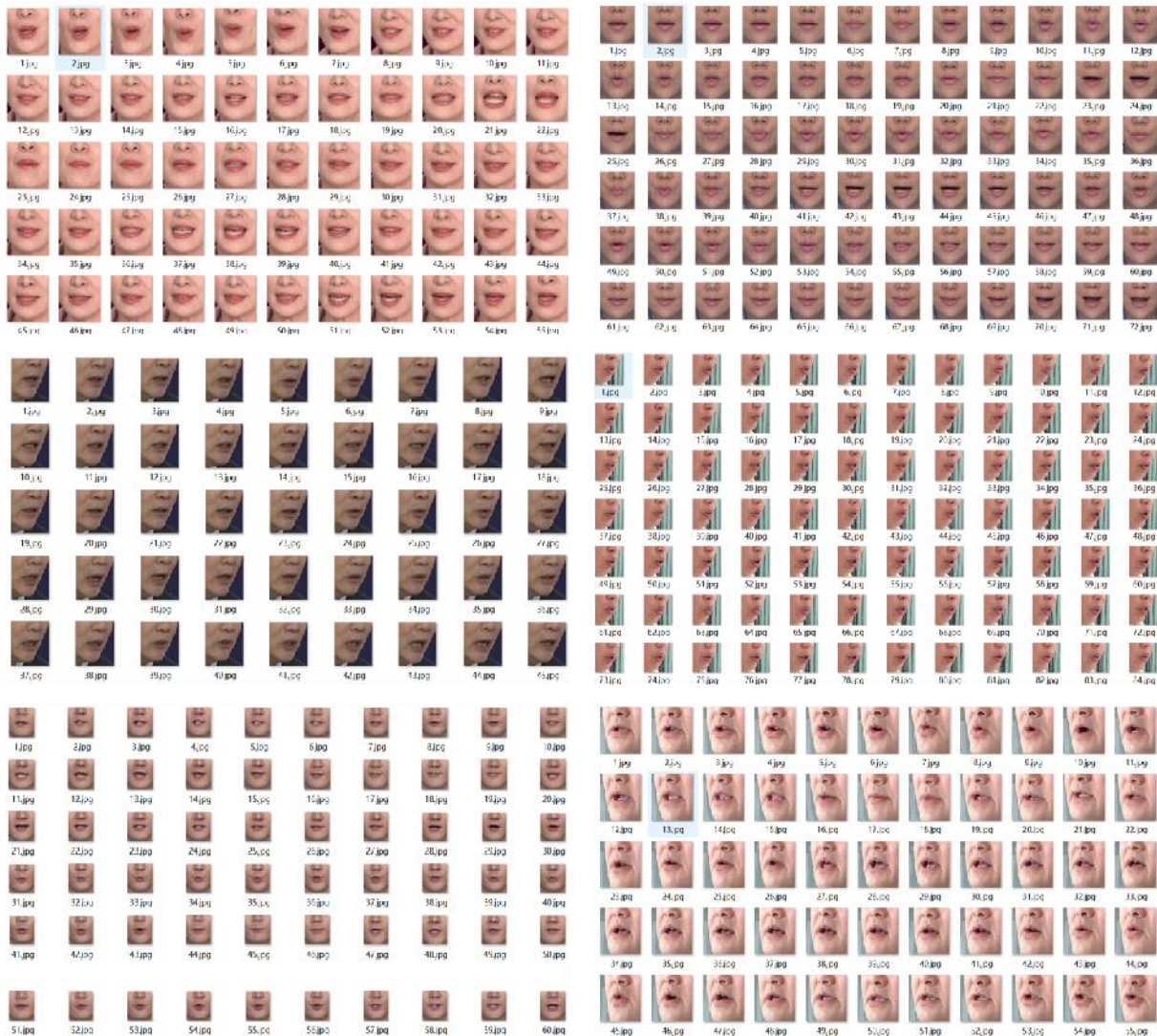


Figure 9 LRW-1000 datasets examples

The advantages of LRW-1000 dataset for this paper are as follows:

(1) This dataset has a large sample size, and the labels used for data processing are provided in English alphabet format, so the range of developers who can use this dataset is large.

(2) As shown in the figure, this dataset is wild and random, which makes it suitable as a dataset for lip-speaking research in complex environments and is in line with the focus of this paper. The dataset encompasses diverse speech conditions, encompassing variations in lighting, video resolution, and speaker attributes such as posture, speech speed, age, gender, and appearance factors like makeup.

(3) The dataset conforms to the reality of word usage by allowing some categories to contain more samples than others. Samples of the same word are not limited to a previously specified range of lengths to allow for different speech rates.

3.2 Model Architecture

In this section, the mathematical models of the three front-end networks are described in detail, laying the groundwork for the simulation experiments and evaluation of the three front-end models, LSTM, LipNet and DenseNet, in the next section, as well as for the proposal and optimization of the novel models in the later section.

3.2.1 LSTM mathematical model

To cope with the problem that RNNs only refer to the previous sequence features, the model in this paper uses a bidirectional RNN network. In the time series classification problem, the time series does not need to have directionality as time passes, as long as the sequence has a logical relationship before and after. Therefore, in time series feature extraction, the logical relationship not only exists in the sequence from front to back, but also exists in the sequence from back to front. Due to the characteristics of the RNN network, the attribute points before and after the sequence do not have completely equal weights, usually the more backward nodes have a greater impact on the output, in order to solve this problem, the usual method is to change the unidirectional network into a bidirectional network, so that the two networks with the same structure and opposite directions jointly determine the output information. So firstly, LSTM was tried.

Long Short-Term Memory Networks (LSTM) are equivalent to a variant of recurrent neural networks, which can solve the problems of gradient vanishing and explosion in traditional RNNs and are particularly suitable for modelling and prediction tasks dealing with long time series data. The core idea of LSTM is to control the flow and storage of information by introducing a gating mechanism to efficiently capture long term dependencies in a time series over a long-time span ^[22]. Unlike traditional RNNs, the LSTM unit contains four main components: an input gate, a forget gate, a cell state, and an output gate. The schematic is as follows in Figure 8 and Figure 9.

At time step (t), the input is x_t , the hidden state is h_t , the cell state is c_t , and the output is y_t .

(1) Input Gate

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (5)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (6)$$

(2) Forget Gate

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (7)$$

(3) Cell Status Updates

$$\tilde{C}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (8)$$

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{C}_t \quad (9)$$

(4) Output Gate

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \quad (10)$$

(5) After obtaining the memory state C_t , the output o_t and hidden state h_t of the current LSTM neural unit can also be calculated as shown below.

$$h_t = o_t \cdot \tanh(c_t) \quad (11)$$

Where σ is a hyperbolic tangent function; i_t, f_t, o_t is the gating vector of the input, forgetting and output gates, which determines the degree of opening and closing of the

corresponding gate; $\tilde{C}_t$ is the candidate cell state, which represents the cell state to be updated; W is the weight matrix and b is the bias term. LSTM definition diagram and frame flow chart are shown in Figure 10 and Figure 11.

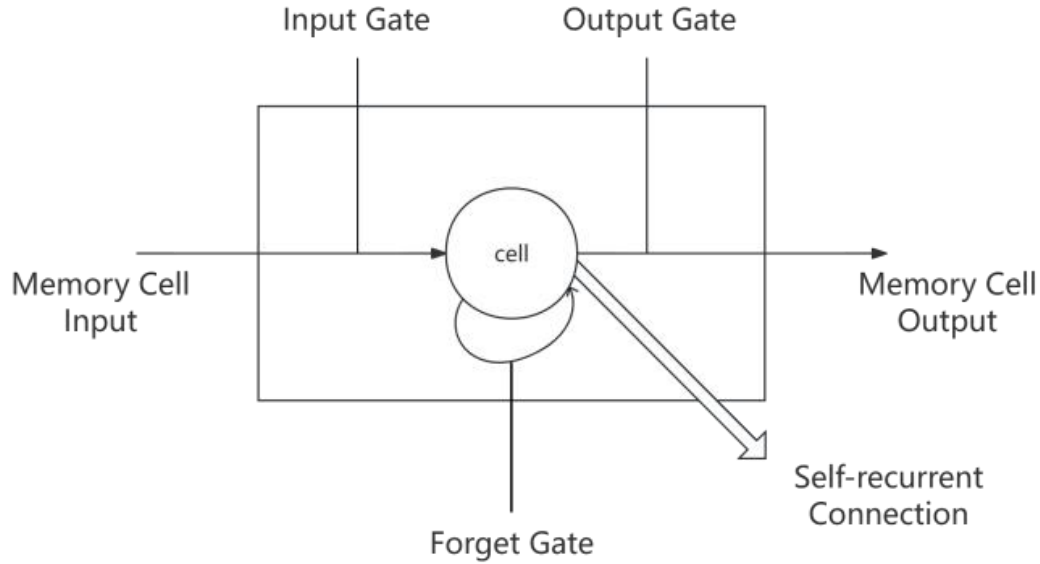


Figure 10 LSTM definition diagram

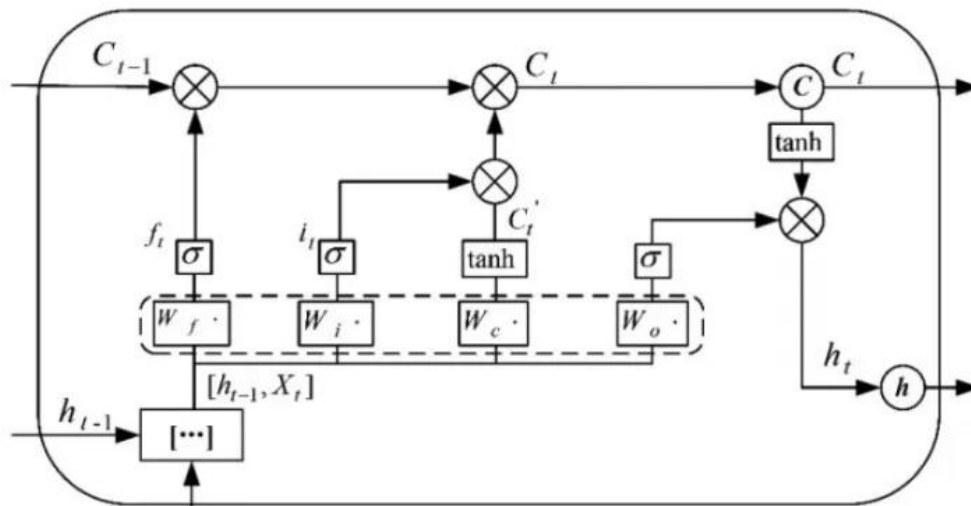


Figure 11 LSTM frame flow chart

3.2.2 LipNet mathematical model

Based on unidirectional LSTM network, this paper thought of LSTM variant - bi-directional LSTM network, as the next model to be investigated, so the LipNet model was found, and it

converged well on the public dataset LRW with high lip recognition rate. Its concrete Architecture is shown in Figure 12.

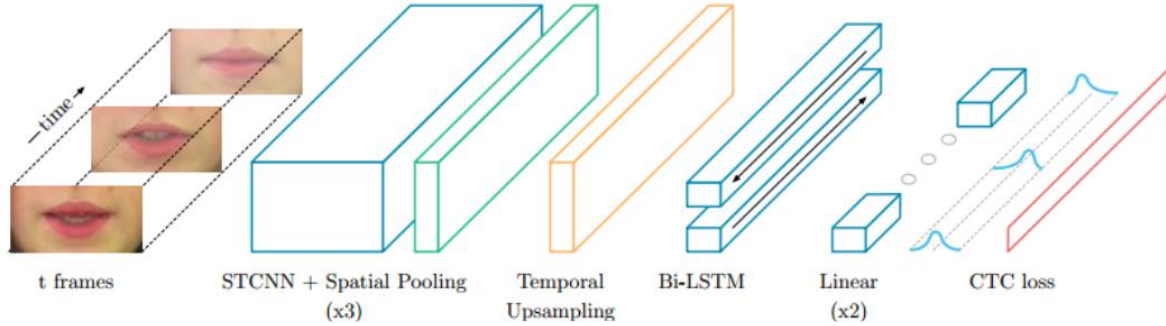


Figure 12 LipNet architecture.

The LipNet model uses T-frame sequences as input and processes the video sequences through 3 layers of STCNN, each layer of STCNN has a spatial maximum pooling layer to reduce the spatial dimensions of the feature maps, and at each timestep, the STCNN extracts the spatio-temporal features by performing convolutional operations on the T-frame sequences. The related mathematical model is as follows:

$$Y = \text{Conv}(X, W) + b \quad (12)$$

where X is the input feature map, W is the convolution kernel, and b is the bias term.

$$Y_{i,j,k} = \sum_{m,n} X_{i+m, j+n, k} \cdot W_{m,n,k} + b_k \quad (13)$$

$$Y_{i,j,k} = \max(X_{i',j',k}) \quad (14)$$

where $Y_{i,j,k}$ is the i, j, k value of the first position of the output feature map and i', j' is the pooling window in the input feature map.

The extracted features are temporarily unsampled and processed by Bi-LSTM; Bi-LSTM is used for sequence modelling of spatio-temporal features extracted by STCNN. A 2-layer feed-forward network and a softmax for final classification are employed to process the output of each time step of the LSTM, allowing it to effectively capture long-term dependencies in sequential data by considering both past and future information. The feedforward network consists of fully connected layers, which include a hidden layer and an output layer. Each hidden layer uses an activation function to introduce nonlinearities, and the final output layer typically uses a softmax function to generate probability distributions for

the categories.

$$h_t^{(1)} = \text{ReLU}(W_1 h_t + b_1) \quad (15)$$

$$h_t^{(2)} = \text{ReLU}(W_2 h_t^{(1)} + b_2) \quad (16)$$

$$\text{Softmax}(y_t) = \frac{e^{W_3 h_t^{(2)} + b_3}}{\sum_k e^{W_3 h_t^{(2)} + b_3}} \quad (17)$$

where ReLU is the modified linear unitary function and W_1, W_2, W_3 and b_1, b_2, b_3 are the corresponding weights and biases.

This end-to-end model is trained using CTC, which compares the LSTM output with the labelled sequences and calculates the loss function for model training. The associated mathematical model is as follows:

$$\text{CTC Loss} = -\sum \pi P(\pi | X) \quad (18)$$

where, $P(\pi | X)$ is the conditional probability of the sequence Y given the input sequence X .

3.2.3 DenseNet mathematical model

This paper uses DenseNet to compare with the previous two models. DenseNet is a densely connected convolutional neural network structure. Compared to the traditional convolutional neural network (CNN), the output of each layer in DenseNet is connected to the inputs of all subsequent layers, forming a densely connected structure. This densely connected design allows features to be more fully propagated and reused in the network, thereby increasing the efficiency and performance of the network while reducing the number of parameters.

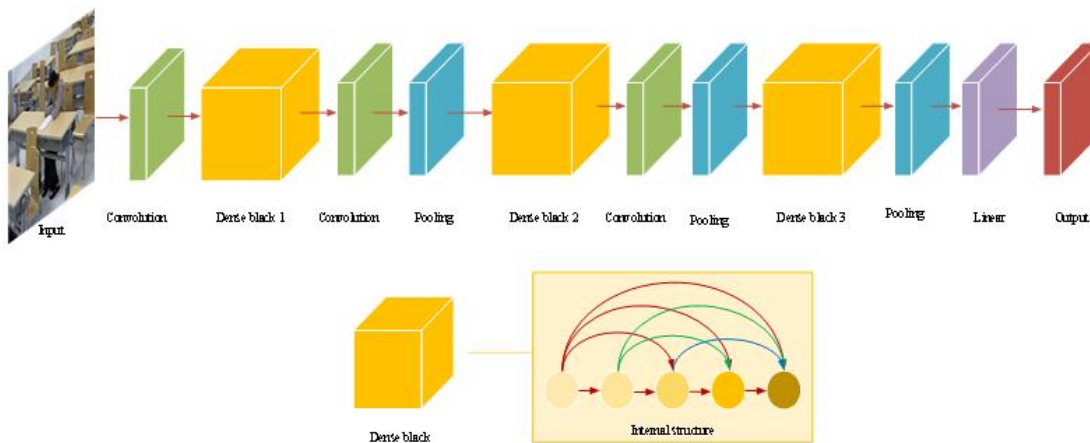


Figure 13 DenseNet architecture.

The output connections of the first 1-1 layers are composed, which is shown in below:

$$X_l = [X_0, X_1, \dots, X_{l-1}] \quad (19)$$

which X_0 is the input image or the output of the previous module.

The network structure of DenseNet mainly consists of DenseBlock and Transition. The dense block consists of several convolutional layers and batch normalization layers, where the output of each convolutional layer is connected to the inputs of all previous layers. This design allows features to exchange a high degree of information within the block. DenseNet model overall architecture is shown in Figure 13 above. The nonlinear combinatorial function $H(\bullet)$ in DenseBlock uses the structure of BN+ReLU+3x3 Conv. Since the inputs of the later layers will be very large, in this paper, bottleneck layer is used inside DenseBlock to reduce the computation, mainly adding 1x1 Conv to the original structure, i.e., BN+ReLU+1x1 Conv+BN+ReLU+3x3 Conv. The structure of DenseNet is shown in Figure 14.

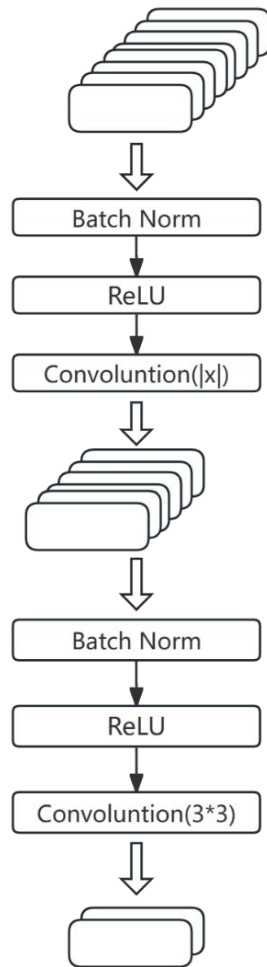


Figure 14 DenseBlock architecture.

If dense block contains L convolutional layers, the output of the first convolutional layer can be expressed as:

$$X_l = H_l([X_0, X_1, \dots, X_{l-1}]) \quad (20)$$

where H_l is the mapping function of the l convolutional layer, and the connection operation is usually performed using stacked convolutional layers (e.g. 3x3 convolution). The operation of each convolutional layer can be expressed as:

$$X_l = H_l([X_0, X_1, \dots, X_{l-1}]) = ReLU(BN(W_l * [X_0, X_1, \dots, X_{l-1}] + b_l)) \quad (21)$$

The transition layer mainly connects two neighboring DenseBlocks and reduces the feature map size, thus reducing the computational complexity of the network. The Transition layer consists of a 1x1 convolution and 2x2 AvgPooling with the structure BN+ReLU+1x1 Conv+2x2 AvgPooling. Assuming that the inputs to the transition layer are X_l , its 1x1 Convolution operation and AvgPooling operation can be expressed as

$$X'_l = BN(X_l) \quad (22)$$

$$X''_l = ReLU(X'_l) \quad (23)$$

$$X'''_l = Conv_{1 \times 1}(X''_l) \quad (24)$$

$$X''''_l = AvgPool_{2 \times 2}(X'''_l) \quad (25)$$

where BN denotes batch normalization, ReLU denotes modified linear unit, $Conv_{1 \times 1}$ denotes 1x1 convolution operation, and $AvgPool_{2 \times 2}$ denotes 2x2 average pooling operation.

4 SIMULATION EXPERIMENTS

This chapter will firstly detail the specific steps and processing details of the experimental design, including the processing of the LRW-1000 dataset, the lip-synthesis information time series transformation algorithm, and a specific description of the experimental methodology. Then the training results of the corresponding models will be presented, and various potential factors in complex environments will be analysed in depth according to different model results. In Section 3, based on the simulation results obtained from the above experiments, a novel optimization scheme is proposed in this paper, aiming to ultimately optimise the model parameters and model structure so as to improve the accuracy of lip recognition.

4.1 Experimental Design

This section will focus on the specific experimental design steps and processing details, including the data preprocessing of LRW-1000, the lipreading information time series transformation algorithm as well as the specific experimental methodology of the simulation experiments. It should be noted that all three front-end networks taken for this simulation experiment are combined with the same structured back-end network containing bidirectional RNNs for the final recognition. The recurrent units used in the experiments are bidirectional gated recurrent units.

4.1.1 Data preprocessing

The pre-processing process of the LRW-1000 video dataset is as follows, including video reading and frame extraction, face positioning and alignment, image pre-processing, ROIs cropping and standardization and data batch processing five parts. The detailed process is described as follows.

(1) Video reading and frame extraction: Read video data from LRW-1000 and extract images frame by frame. Perform data cleaning on each frame to remove possible noise or abnormal frames. The processing results of videos are shown in Figure 15.

(2) Face positioning and alignment: Firstly, face detection is performed using SeetaFaceEngine2 toolkit and face key points are extracted and through the face keypoints, the face in each frame is aligned so that the eyes are almost on the horizontal line. According to the position of the mouth key points, the square mouth-centered regions (ROIs) are

extracted. The specific implementation is as follows: Common face key point detection techniques are able to extract 68 face key points as well as enough to give a more accurate description of the face profile.

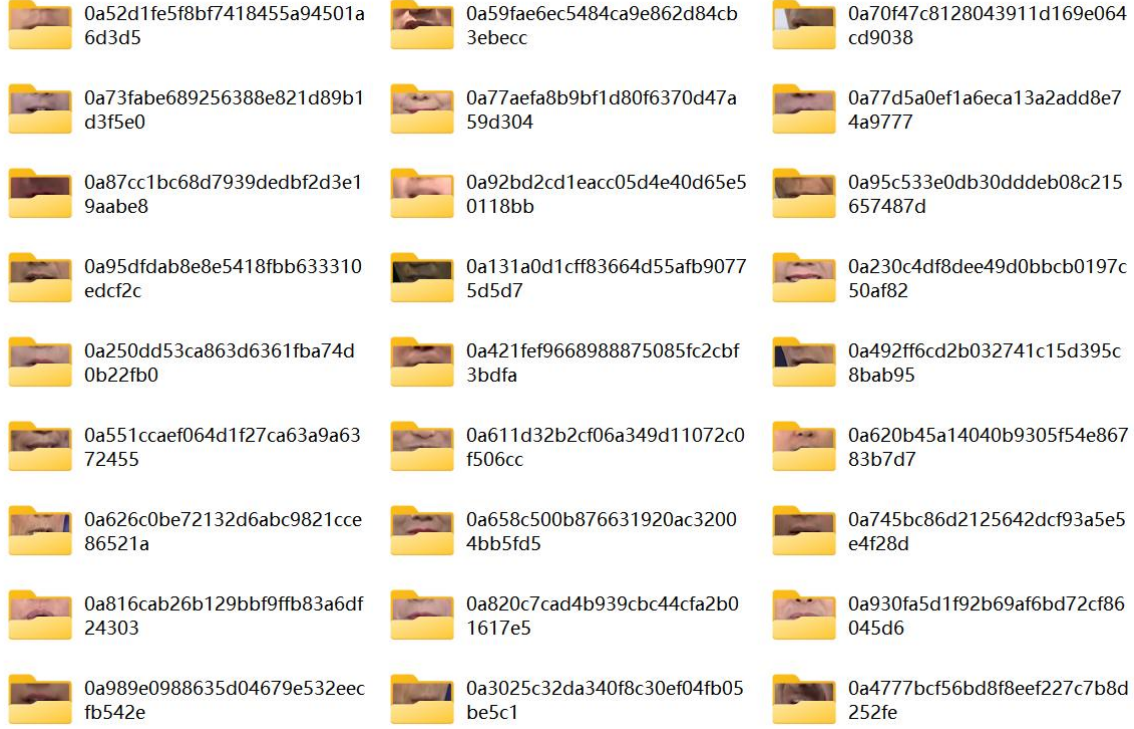


Figure 15 Data frame processing

In the lip recognition method, it is not necessary to consider all the face key points, but only need to focus on the 20 key points that represent the lip features, i.e., key points No. 48 to No. 67. Among them, a total of 12 key points from No. 48 to No. 59 extract information about the outer contour of the lips, and a total of 8 key points from No. 60 to No. 67 extract information about the inner contour of the lips. After extracting the key point information, the perimeter $C1$ of the outer lip contour is calculated with the area $S1$, and the length-to-width ratio $P1$, while the perimeter $C2$ of the inner lip contour is calculated with the area $S2$, and the length-to-width ratio $P2$. The 20 key point coordinate information is normalized with the calculated 6 contour feature information to obtain a 46-dimensional time series. In face key point detection techniques, a classical way ^[23] to achieve this function is to use a cascading regression tree. In the first level of regression, the iterative process is shown in Equation(26).

$$\hat{S}^{t+1} = \hat{S}^t + \gamma_t(I, \hat{S}^t) \quad (26)$$

where $\hat{S}^t$ is the position of the key point of the face at the t^{th} round of regression, I is the training set image, and γ_t is the regressor of the second layer. $\hat{S}^t$ and I are inputted into the regressor γ_t to get the update amount of each key point position, which is accumulated to the key point position $\hat{S}^t$ at the t round of regression to get the key point position $\hat{S}^{t+1}$ at the $t+1^{th}$ round of regression. The second layer regressor γ_t consists of a series of regression trees constructed by using the Boosting additive model. Since the current face key point detection techniques can already meet the needs of the method in this chapter, the method in this chapter does not make any improvement to them, but directly adopts the pre-trained model in the machine learning library Dlib for face key point detection. The specific information of the 68 key points is shown in the following Figure 16.

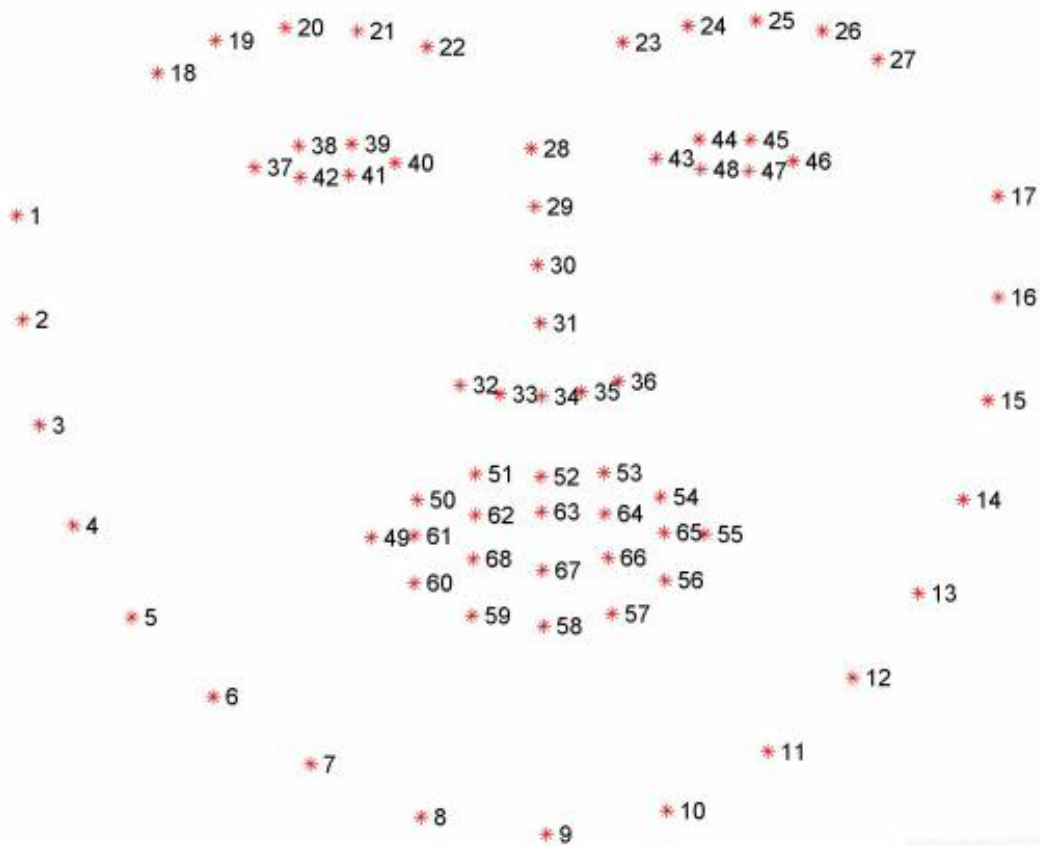


Figure 16 Face landmarks

(3) Image pre-processing: The extracted ROIs were resized to a fixed size of $122 * 122$ pixels. Then the ROIs were mirror flipped to expand the data and denoised by applying bilateral filtering with 60% probability. In other words, the size of the ROI bounding box is determined by:

$$w = \min\{3.2d_{AB}, \max\{2d_{AB}, 1.12x_M - 0.88x_N\}\} \quad (27)$$

Where x_M and x_N , are the x-coordinates of the left and right corners of the mouth. d_{AB} is twice the distance from the center of the speaker's nose to the very center.

(4) ROIs cropping and standardization: The size of ROIs was determined based on the distance between the corners of the mouth or the distance from the tip of the nose to the centre of the mouth, and the rotation, center coordinates and size were smoothed by applying a first order Savitzky-Golay filter.

(5) Data batch processing: To maintain a uniform number of frames for the samples in each batch, frame padding is required for each batch. The number of frames for each sample in the batch is fixed to the maximum number of frames in the sample, thus ensuring consistency in the network input.

4.1.2 Implementation of lip sequence conversion algorithm

After obtaining the coordinates of the lip keypoints, how to convert them into a multidimensional time series is the difficulty of the research. The algorithm used in this chapter refers to the gait recognition method based on multidimensional time series [21] [24], which converts the two-dimensional coordinate information of the key point into one-dimensional angle information, which is used as the core dimension of the lip multidimensional time series. In addition to the fact that angle is easier to handle than 2D coordinates in terms of data structure, because it is difficult to ensure that the face always faces the camera in a completely horizontal face view during video acquisition, it is inevitable that there will be problems such as lip displacement and angular tilt between the front and back frames, and the LRW-1000 dataset is extremely wild, so it is necessary to additionally carry out the front and back frame alignment process when using coordinate information, which increases the complexity of the algorithm construction. These problems can be avoided by using angular information, because the key feature of lip-speaking lies in the change of lip shape during speech, which is reflected by the change of angle between the key points of the lips.

For any key point K of the lips, there must exist two neighboring points M and N on both sides of it. with the horizontal and vertical coordinates of the three points known, the specific calculation process of the angle is shown in the following formula [36].

$$\begin{cases} \theta_{K1} = \arctan \frac{y_K - y_M}{x_K - x_M} \\ \theta_{K2} = \arctan \frac{y_K - y_N}{x_K - x_N} \\ \angle MKN = \theta_K = \begin{cases} |\theta_{K1} - \theta_{K2}| & \text{if } \theta_{K1} \times \theta_{K2} \geq 0 \\ |\theta_{K1}| + |\theta_{K2}| & \text{if } \theta_{K1} \times \theta_{K2} < 0 \end{cases} \end{cases} \quad (28)$$

In addition to the angular features, the lip contour distance features are also an indispensable part. The method in this chapter captures the height h_0 and width w_0 of the outer lip contour, the height h_j and width w_j of the inner contour, a total of four contour features. Together with the aforementioned key point angle information, the complete set of lip features $T = \{\theta_1, \theta_2, \dots, \theta_n, w_o, w_j, h_o, h_j\}$, each element in the set describes the features of a certain frame of an image in the lip recognition process in a certain dimension, and by connecting the extracted feature values in each frame, a multidimensional lip time sequence with sequence length as the number of lip video frames and the number of dimensions as the number of elements in the feature set is obtained, which also completes the conversion from lip key point coordinates to lip sequences.

4.1.3 Experimental methods

In this paper, the experiments in this paper use the LRW-1000 dataset, which is divided into three parts: the Training Set, the Test Set, and the Validation Set. The video duration ratio of these three parts is about 8:1:1 to ensure that few speakers overlap in the classification of the three categories. Specifically, the Training Set contains 634 videos, the Test Set contains 82 videos, and the Validation Set contains 122 videos. Each time series in the training set is cropped at the same random position before inputting into the model, while the centralised cropping method is used for validation and testing.

To speed up the training process, in this paper we divide the training into two phases. In the first phase, the experiment focuses on shorter sequences with lengths less than 30, which increases the amount of training data. When the model shows a convergence trend, the remaining sequences are added to the training set. This paper also employs random repetition of samples throughout the training process to further accelerate the convergence rate.

It should be noted that all three front-end networks taken for this simulation experiment are combined with the same structured back-end network containing bidirectional RNNs for

the final recognition. The recurrent units used in the experiments are bidirectional gated recurrent units.

4.2 Results Comparison

This section will show the comparison of training results of LSTM, DenseNet models on LRW-1000 dataset.

4.2.1 LSTM model

The results of the loss rate comparison between the training set, test set and validation set at different time periods are as follows. Figure 17 shows that as training round increases, the Batch loss of the training set, test set and validation set all show a gradual decrease. In the initial stage, the loss rate decreases more rapidly, but as the training proceeds, the decrease gradually slows down and eventually stabilizes. This phenomenon can be attributed to the model's weak ability to fit the data in the initial stage, and as training proceeds, the model gradually learns the features of the lip-speaking data and optimizes them, and the loss gradually decreases. However, as the training progresses, the model's learning ability becomes saturated, resulting in a slower decrease in the loss rate, which eventually stabilizes.

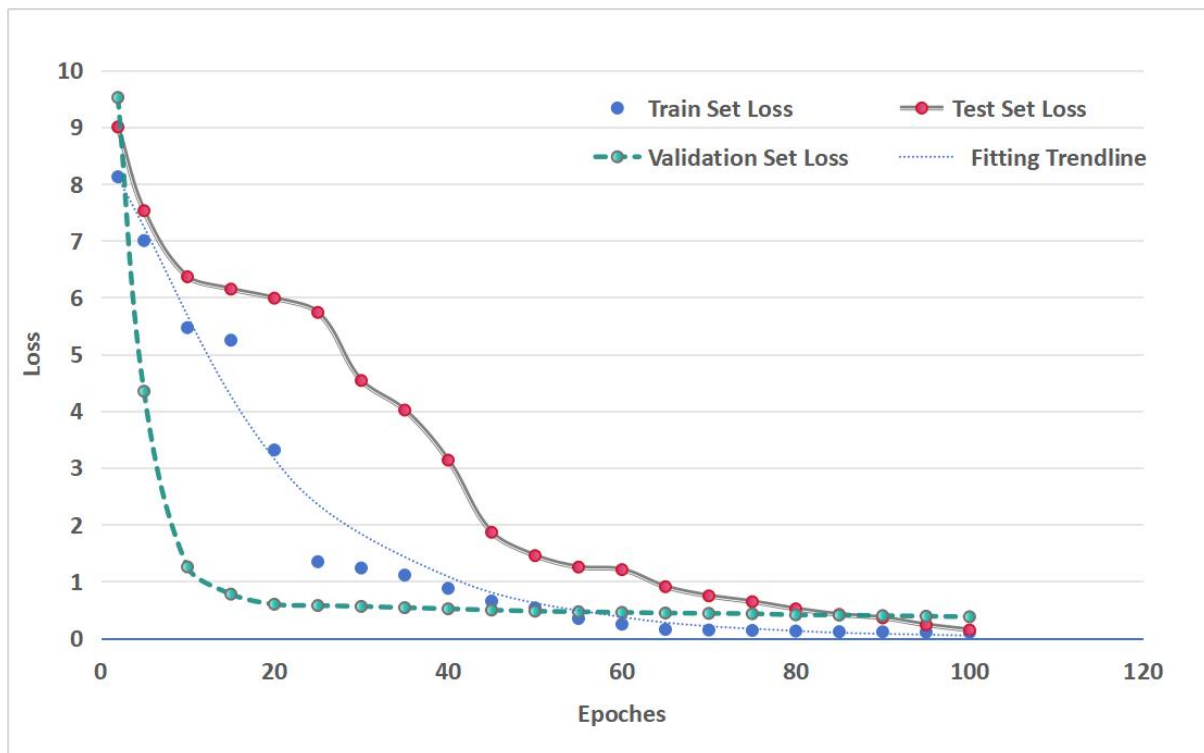


Figure 17 Comparison of loss rates of train Set, test Set, validation set under different epochs

Figure 18 shows the recognition accuracy results for the training set, test set and validation set at different epochs. According to the data, the recognition rate of the training set is about 45.04075%, the recognition rate of the test set is about 33.7778996%, while the accuracy of the validation set is around 21.45462%. The low recognition accuracy of the model under the LRW-1000 dataset indicates the limitations of the model in handling these complex situations. This suggests that further optimisation of the model structure is required to improve the performance of the model.

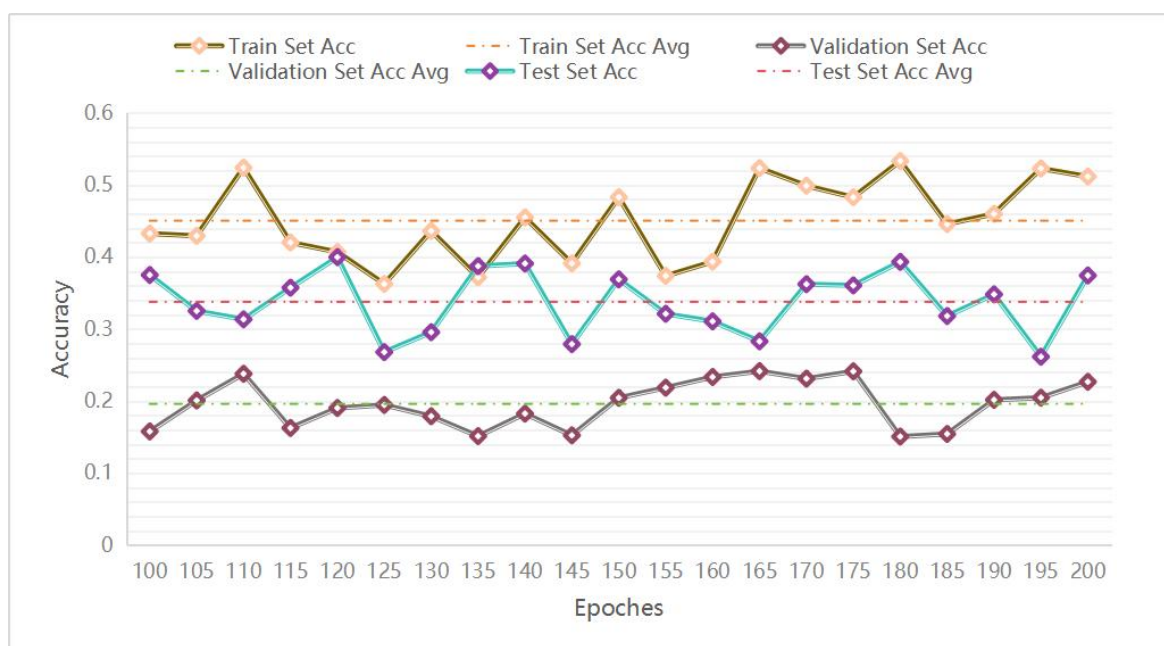


Figure 18 Recognition accuracy results of train set, test set, validation set under different epochs

4.2.2 DenseNet model

The figure below first shows the DenseNet model data training test process. It shows the data volumes, accuracy batch, estimated time, batch loss, total accuracy, etc.

```
train from scratch
index file: E:\LRW1000_Public_packed\info\trn_1000.txt
num of pinyins: 1000
num of data: 603193
max video length 57
index file: E:\LRW1000_Public_packed\info\val_1000.txt
num of pinyins: 1000
num of data: 63237
max video length 51
index file: E:\LRW1000_Public_packed\info\tst_1000.txt
num of pinyins: 1000
num of data: 51588
max video length 35
```

(a) Total training volume

Process: [11/ 90 (11%)]	Acc batch:1.0000	Acc total:0.0909	Cost time: 3s	Estimated time: 21s
Process: [21/ 90 (22%)]	Acc batch:1.0000	Acc total:0.2857	Cost time: 7s	Estimated time: 26s
Process: [31/ 90 (34%)]	Acc batch:0.0000	Acc total:0.2903	Cost time: 12s	Estimated time: 23s
Process: [41/ 90 (45%)]	Acc batch:0.0000	Acc total:0.2195	Cost time: 16s	Estimated time: 20s
Process: [51/ 90 (56%)]	Acc batch:0.0000	Acc total:0.1765	Cost time: 21s	Estimated time: 17s
Process: [61/ 90 (67%)]	Acc batch:0.0000	Acc total:0.1475	Cost time: 23s	Estimated time: 11s
Process: [71/ 90 (79%)]	Acc batch:1.0000	Acc total:0.1831	Cost time: 28s	Estimated time: 7s
Process: [81/ 90 (90%)]	Acc batch:1.0000	Acc total:0.2840	Cost time: 31s	Estimated time: 4s
Process: [90/ 90 (100%)]	Acc batch:0.0000	Acc total:0.2667	Cost time: 35s	Estimated time: 0s
tst Epoch: 0	Acc:0.2667			

(b) Test set results

Process: [11/ 619 (2%)]	Loss batch: 6.9389	Loss total: 6.7936	Acc batch:0.0000	Acc total:0.0000	Cost time: 3s	Estimated time: 157s
Process: [21/ 619 (3%)]	Loss batch: 5.7553	Loss total: 6.3387	Acc batch:0.0000	Acc total:0.1429	Cost time: 6s	Estimated time: 194s
Process: [31/ 619 (5%)]	Loss batch: 4.4231	Loss total: 5.8703	Acc batch:0.0000	Acc total:0.1935	Cost time: 11s	Estimated time: 207s
Process: [41/ 619 (6%)]	Loss batch: 4.3323	Loss total: 5.3915	Acc batch:0.0000	Acc total:0.2683	Cost time: 14s	Estimated time: 201s
Process: [51/ 619 (8%)]	Loss batch: 0.9825	Loss total: 5.0700	Acc batch:1.0000	Acc total:0.2745	Cost time: 17s	Estimated time: 196s
Process: [61/ 619 (10%)]	Loss batch: 2.7296	Loss total: 4.9468	Acc batch:0.0000	Acc total:0.2623	Cost time: 21s	Estimated time: 193s
Process: [71/ 619 (11%)]	Loss batch: 2.6990	Loss total: 4.7051	Acc batch:0.0000	Acc total:0.2394	Cost time: 25s	Estimated time: 194s
Process: [81/ 619 (13%)]	Loss batch: 3.6764	Loss total: 4.5408	Acc batch:0.0000	Acc total:0.2593	Cost time: 28s	Estimated time: 189s
Process: [91/ 619 (15%)]	Loss batch: 2.3956	Loss total: 4.2837	Acc batch:0.0000	Acc total:0.2747	Cost time: 31s	Estimated time: 184s
Process: [101/ 619 (16%)]	Loss batch: 1.1381	Loss total: 4.1337	Acc batch:1.0000	Acc total:0.2970	Cost time: 35s	Estimated time: 179s
Process: [111/ 619 (18%)]	Loss batch: 2.1944	Loss total: 4.0216	Acc batch:0.0000	Acc total:0.2883	Cost time: 39s	Estimated time: 182s
Process: [121/ 619 (19%)]	Loss batch: 1.4789	Loss total: 3.8394	Acc batch:0.0000	Acc total:0.2975	Cost time: 43s	Estimated time: 179s
Process: [131/ 619 (21%)]	Loss batch: 0.7773	Loss total: 3.7401	Acc batch:1.0000	Acc total:0.2977	Cost time: 46s	Estimated time: 174s
Process: [141/ 619 (23%)]	Loss batch: 1.1072	Loss total: 3.6208	Acc batch:1.0000	Acc total:0.3050	Cost time: 51s	Estimated time: 174s
Process: [151/ 619 (24%)]	Loss batch: 1.1864	Loss total: 3.4933	Acc batch:0.0000	Acc total:0.3113	Cost time: 57s	Estimated time: 178s
Process: [161/ 619 (26%)]	Loss batch: 2.1857	Loss total: 3.4173	Acc batch:0.0000	Acc total:0.3168	Cost time: 62s	Estimated time: 178s
Process: [171/ 619 (28%)]	Loss batch: 1.3796	Loss total: 3.3221	Acc batch:1.0000	Acc total:0.3158	Cost time: 65s	Estimated time: 172s
Process: [181/ 619 (29%)]	Loss batch: 1.7042	Loss total: 3.2455	Acc batch:0.0000	Acc total:0.3204	Cost time: 72s	Estimated time: 175s
Process: [191/ 619 (31%)]	Loss batch: 2.2404	Loss total: 3.1607	Acc batch:0.0000	Acc total:0.3246	Cost time: 77s	Estimated time: 173s
Process: [201/ 619 (32%)]	Loss batch: 1.0574	Loss total: 3.0820	Acc batch:1.0000	Acc total:0.3184	Cost time: 81s	Estimated time: 169s
Process: [211/ 619 (34%)]	Loss batch: 1.2336	Loss total: 2.9971	Acc batch:0.0000	Acc total:0.3223	Cost time: 84s	Estimated time: 163s
Process: [221/ 619 (36%)]	Loss batch: 4.1704	Loss total: 2.9557	Acc batch:0.0000	Acc total:0.3167	Cost time: 87s	Estimated time: 158s
Process: [231/ 619 (37%)]	Loss batch: 1.2436	Loss total: 2.8994	Acc batch:0.0000	Acc total:0.3203	Cost time: 92s	Estimated time: 154s
Process: [241/ 619 (39%)]	Loss batch: 2.0786	Loss total: 2.8523	Acc batch:0.0000	Acc total:0.3112	Cost time: 95s	Estimated time: 149s
Process: [251/ 619 (40%)]	Loss batch: 1.3287	Loss total: 2.8031	Acc batch:0.0000	Acc total:0.3028	Cost time: 98s	Estimated time: 144s
Process: [261/ 619 (42%)]	Loss batch: 0.5478	Loss total: 2.7517	Acc batch:1.0000	Acc total:0.2989	Cost time: 102s	Estimated time: 140s
Process: [271/ 619 (44%)]	Loss batch: 3.0145	Loss total: 2.7193	Acc batch:0.0000	Acc total:0.2989	Cost time: 105s	Estimated time: 135s
Process: [281/ 619 (45%)]	Loss batch: 1.8517	Loss total: 2.6907	Acc batch:0.0000	Acc total:0.2883	Cost time: 110s	Estimated time: 133s
Process: [291/ 619 (47%)]	Loss batch: 2.1302	Loss total: 2.6679	Acc batch:0.0000	Acc total:0.2818	Cost time: 114s	Estimated time: 128s
Process: [301/ 619 (49%)]	Loss batch: 2.5845	Loss total: 2.6267	Acc batch:0.0000	Acc total:0.2857	Cost time: 119s	Estimated time: 126s
Process: [311/ 619 (50%)]	Loss batch: 3.2267	Loss total: 2.6010	Acc batch:0.0000	Acc total:0.2830	Cost time: 125s	Estimated time: 124s
Process: [321/ 619 (52%)]	Loss batch: 0.8376	Loss total: 2.5589	Acc batch:1.0000	Acc total:0.2897	Cost time: 130s	Estimated time: 121s
Process: [331/ 619 (53%)]	Loss batch: 1.0079	Loss total: 2.5239	Acc batch:1.0000	Acc total:0.2991	Cost time: 133s	Estimated time: 116s
Process: [341/ 619 (55%)]	Loss batch: 0.6845	Loss total: 2.5099	Acc batch:1.0000	Acc total:0.2962	Cost time: 138s	Estimated time: 113s
Process: [351/ 619 (57%)]	Loss batch: 1.1840	Loss total: 2.5047	Acc batch:1.0000	Acc total:0.2963	Cost time: 142s	Estimated time: 108s
Process: [361/ 619 (58%)]	Loss batch: 0.5806	Loss total: 2.4815	Acc batch:1.0000	Acc total:0.2964	Cost time: 146s	Estimated time: 104s
Process: [371/ 619 (60%)]	Loss batch: 1.8540	Loss total: 2.4551	Acc batch:0.0000	Acc total:0.2992	Cost time: 150s	Estimated time: 100s
Process: [381/ 619 (61%)]	Loss batch: 1.3447	Loss total: 2.4313	Acc batch:0.0000	Acc total:0.3045	Cost time: 152s	Estimated time: 95s
Process: [391/ 619 (63%)]	Loss batch: 3.0383	Loss total: 2.4162	Acc batch:0.0000	Acc total:0.2992	Cost time: 158s	Estimated time: 92s
Process: [401/ 619 (65%)]	Loss batch: 1.5098	Loss total: 2.3990	Acc batch:0.0000	Acc total:0.2968	Cost time: 162s	Estimated time: 89s
Process: [411/ 619 (66%)]	Loss batch: 0.6207	Loss total: 2.3832	Acc batch:1.0000	Acc total:0.2993	Cost time: 166s	Estimated time: 84s
Process: [421/ 619 (68%)]	Loss batch: 2.0148	Loss total: 2.3633	Acc batch:0.0000	Acc total:0.3040	Cost time: 171s	Estimated time: 81s
Process: [431/ 619 (70%)]	Loss batch: 1.3100	Loss total: 2.3520	Acc batch:0.0000	Acc total:0.2970	Cost time: 175s	Estimated time: 77s
Process: [441/ 619 (71%)]	Loss batch: 1.3233	Loss total: 2.3416	Acc batch:1.0000	Acc total:0.2948	Cost time: 179s	Estimated time: 72s
Process: [451/ 619 (73%)]	Loss batch: 1.6397	Loss total: 2.3254	Acc batch:0.0000	Acc total:0.2927	Cost time: 182s	Estimated time: 68s
Process: [461/ 619 (74%)]	Loss batch: 0.5545	Loss total: 2.3039	Acc batch:1.0000	Acc total:0.2950	Cost time: 186s	Estimated time: 64s
Process: [471/ 619 (76%)]	Loss batch: 1.0036	Loss total: 2.2899	Acc batch:1.0000	Acc total:0.2930	Cost time: 190s	Estimated time: 60s
Process: [481/ 619 (78%)]	Loss batch: 1.3724	Loss total: 2.2736	Acc batch:0.0000	Acc total:0.2911	Cost time: 203s	Estimated time: 58s
Process: [491/ 619 (79%)]	Loss batch: 1.0434	Loss total: 2.2640	Acc batch:1.0000	Acc total:0.2872	Cost time: 206s	Estimated time: 54s
Process: [501/ 619 (81%)]	Loss batch: 1.4375	Loss total: 2.2471	Acc batch:0.0000	Acc total:0.2874	Cost time: 211s	Estimated time: 50s
Process: [511/ 619 (83%)]	Loss batch: 1.7533	Loss total: 2.2295	Acc batch:0.0000	Acc total:0.2916	Cost time: 216s	Estimated time: 46s
Process: [521/ 619 (84%)]	Loss batch: 1.2890	Loss total: 2.2178	Acc batch:0.0000	Acc total:0.2917	Cost time: 221s	Estimated time: 42s
Process: [531/ 619 (86%)]	Loss batch: 2.6279	Loss total: 2.2034	Acc batch:0.0000	Acc total:0.2938	Cost time: 228s	Estimated time: 38s
Process: [541/ 619 (87%)]	Loss batch: 3.0295	Loss total: 2.2002	Acc batch:0.0000	Acc total:0.2921	Cost time: 230s	Estimated time: 33s
Process: [551/ 619 (89%)]	Loss batch: 2.2081	Loss total: 2.1902	Acc batch:0.0000	Acc total:0.2922	Cost time: 234s	Estimated time: 29s
Process: [561/ 619 (91%)]	Loss batch: 1.1557	Loss total: 2.1844	Acc batch:0.0000	Acc total:0.2888	Cost time: 238s	Estimated time: 25s
Process: [571/ 619 (92%)]	Loss batch: 0.7093	Loss total: 2.1723	Acc batch:1.0000	Acc total:0.2890	Cost time: 243s	Estimated time: 20s
Process: [581/ 619 (94%)]	Loss batch: 1.4028	Loss total: 2.1541	Acc batch:0.0000	Acc total:0.2926	Cost time: 247s	Estimated time: 16s
Process: [591/ 619 (95%)]	Loss batch: 1.7665	Loss total: 2.1377	Acc batch:0.0000	Acc total:0.2961	Cost time: 255s	Estimated time: 12s
Process: [601/ 619 (97%)]	Loss batch: 2.0876	Loss total: 2.1261	Acc batch:0.0000	Acc total:0.2962	Cost time: 259s	Estimated time: 8s
Process: [611/ 619 (99%)]	Loss batch: 0.9658	Loss total: 2.1295	Acc batch:1.0000	Acc total:0.2962	Cost time: 264s	Estimated time: 3s
Process: [619/ 619 (100%)]	Loss batch: 1.6033	Loss total: 2.1221	Acc batch:0.0000	Acc total:0.2973	Cost time: 267s	Estimated time: 0s
train Epoch: 0	Loss: 2.1221	Acc:0.2973				

(c) Training set results

Process: [11/ 194 (5%)]	Acc batch:0.0000	Acc total:0.0000	Cost time: 9s	Estimated time: 162s
Process: [21/ 194 (10%)]	Acc batch:0.0000	Acc total:0.0000	Cost time: 14s	Estimated time: 119s
Process: [31/ 194 (16%)]	Acc batch:1.0000	Acc total:0.0645	Cost time: 17s	Estimated time: 91s
Process: [41/ 194 (21%)]	Acc batch:0.0000	Acc total:0.0976	Cost time: 19s	Estimated time: 74s
Process: [51/ 194 (26%)]	Acc batch:0.0000	Acc total:0.0784	Cost time: 22s	Estimated time: 63s
Process: [61/ 194 (31%)]	Acc batch:0.0000	Acc total:0.0656	Cost time: 24s	Estimated time: 53s
Process: [71/ 194 (36%)]	Acc batch:0.0000	Acc total:0.0563	Cost time: 27s	Estimated time: 47s
Process: [81/ 194 (41%)]	Acc batch:0.0000	Acc total:0.0494	Cost time: 29s	Estimated time: 41s
Process: [91/ 194 (47%)]	Acc batch:0.0000	Acc total:0.0440	Cost time: 31s	Estimated time: 35s
Process: [101/ 194 (52%)]	Acc batch:0.0000	Acc total:0.0396	Cost time: 33s	Estimated time: 30s
Process: [111/ 194 (57%)]	Acc batch:0.0000	Acc total:0.0360	Cost time: 35s	Estimated time: 27s
Process: [121/ 194 (62%)]	Acc batch:0.0000	Acc total:0.0331	Cost time: 39s	Estimated time: 24s
Process: [131/ 194 (67%)]	Acc batch:0.0000	Acc total:0.0305	Cost time: 42s	Estimated time: 20s
Process: [141/ 194 (73%)]	Acc batch:0.0000	Acc total:0.0284	Cost time: 44s	Estimated time: 17s
Process: [151/ 194 (78%)]	Acc batch:0.0000	Acc total:0.0265	Cost time: 47s	Estimated time: 13s
Process: [161/ 194 (83%)]	Acc batch:0.0000	Acc total:0.0248	Cost time: 52s	Estimated time: 11s
Process: [171/ 194 (88%)]	Acc batch:0.0000	Acc total:0.0234	Cost time: 62s	Estimated time: 8s
Process: [181/ 194 (93%)]	Acc batch:1.0000	Acc total:0.0773	Cost time: 66s	Estimated time: 5s
Process: [191/ 194 (98%)]	Acc batch:0.0000	Acc total:0.1099	Cost time: 70s	Estimated time: 1s
Process: [194/ 194 (100%)]	Acc batch:0.0000	Acc total:0.1082	Cost time: 71s	Estimated time: 0s
val Epoch: 0	Acc:0.1082			

(d) Validation set results

Figure 19 DenseNet model data training testing process

Based on the above training results, Figure 20 illustrates the loss rates of the training set, test set and validation set under different epochs. Observing the graphs, it can be seen that the overall batch loss for the three classification sets shows an unstable decreasing trend, indicating that the model has the potential to learn.

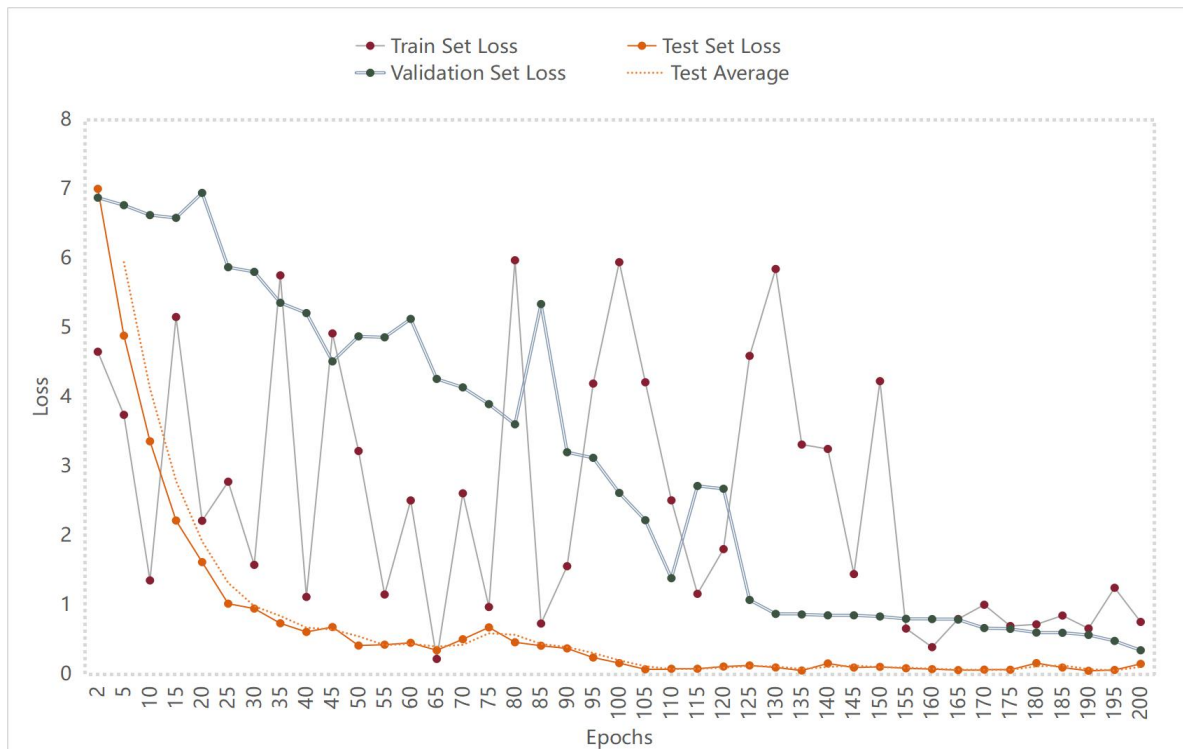


Figure 20 Comparison of batch loss of train set, test set, validation set under different epochs

Parameter tuning and optimization are continuously performed during the training process based on the input data. In the initial stage the model has a weak ability to fit the data and therefore the loss rate fluctuates more dramatically. However, as training proceeds, the model gradually learns the features and patterns of the data, and the loss rate gradually decreases. Although the overall trend is decreasing, the process of decreasing loss rate fluctuates due to the randomness that may exist in the training process. It verifies that the model has certain learning ability and generalization ability.

Figure 21 shows the comparison results of the recognition rates of the three classification sets in different rounds. Observing the graph, it can be seen that as the number of training rounds increases, the recognition rates of the DenseNet model in the three classification sets, except for a little fluctuation, show an overall trend of gradual increase. This phenomenon is mainly due to the fact that the DenseNet model has a strong feature learning and characterization ability when dealing with complex datasets and is able to better capture the higher-order feature relationships between the data, thus improving the recognition performance of the model. In addition, DenseNet's dense connectivity and deep feature transfer mechanism in the network structure also help to improve the model's learning and generalization ability. This result implies that DenseNet exhibits better learning ability and adaptability compared to LSTM on this dataset.

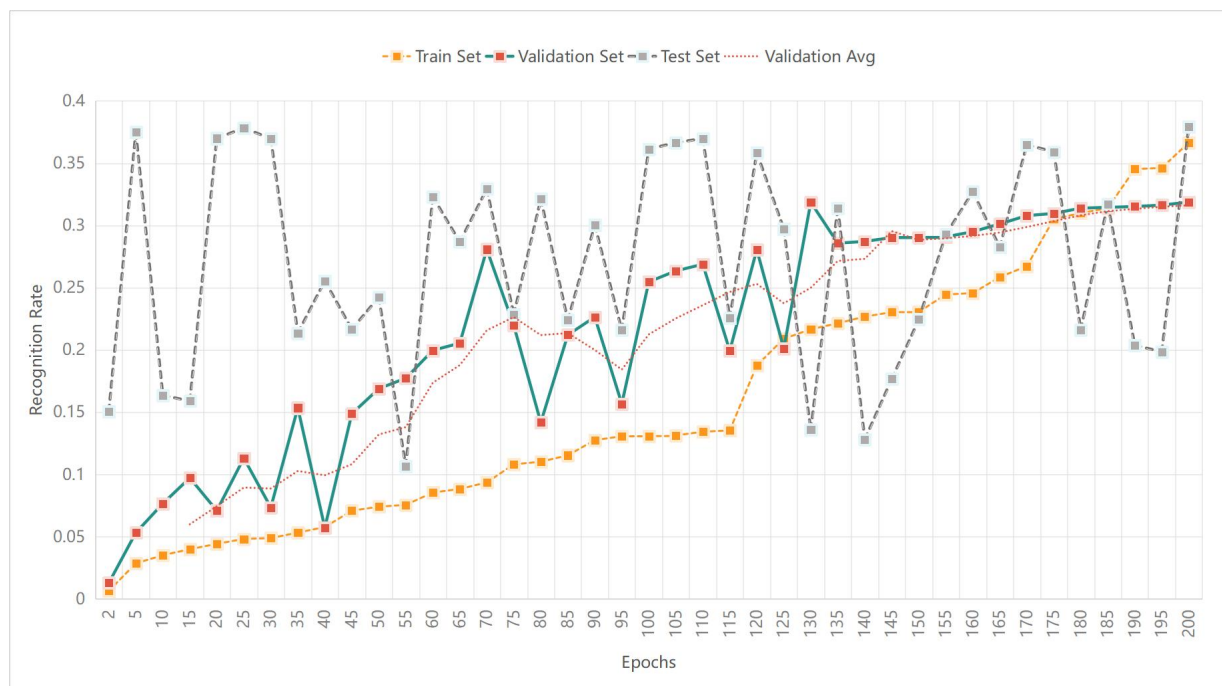


Figure 21 DenseNet model recognition rate results

4.2.3 Model comparison

From the moving average trend lines of the two models in the following Figure 22, it can be observed that when the Epochs are greater than 100, the recognition rate of the DenseNet model shows a tendency to stabilize, while the curve of the LSTM model starts from epoch=0 to show an overall irregular characteristic. This phenomenon may be since DenseNet is more adapted to the characteristics of the dataset in terms of model structure and parameter settings, which enables it to stabilize at a relatively high recognition rate level in the later stages of training. In contrast, the LSTM model may have been limited by its more complex structure, resulting in a more unstable change in recognition rate.

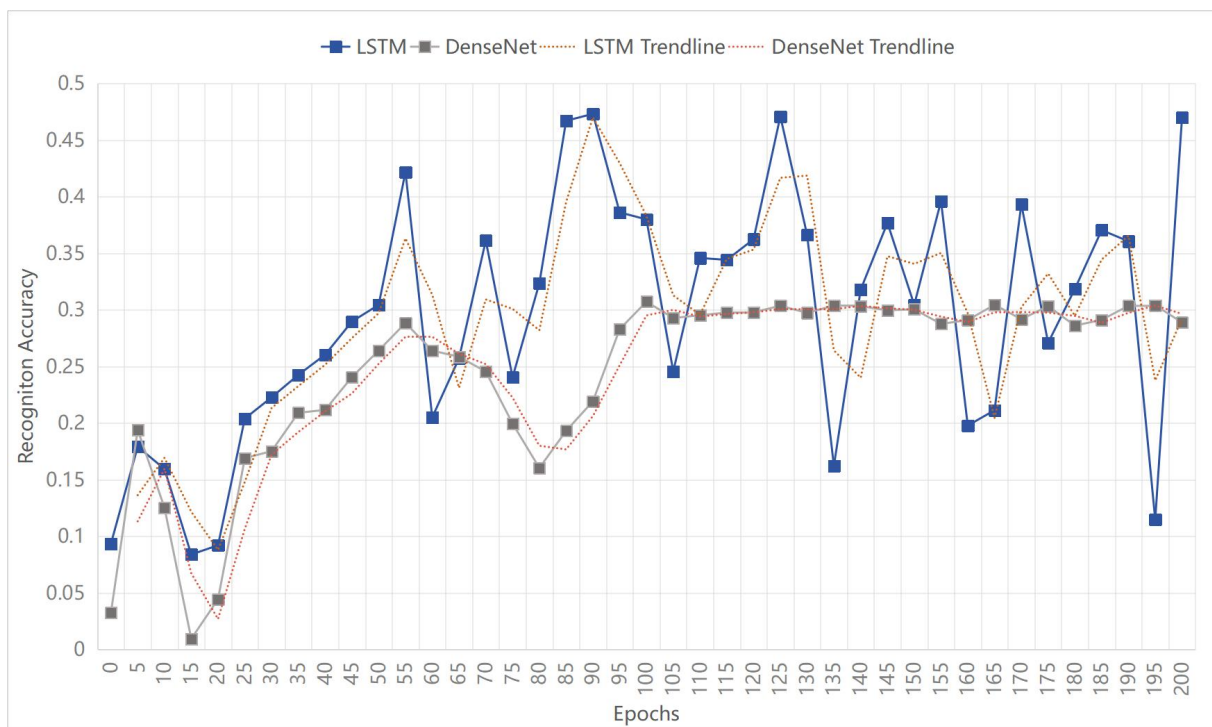


Figure 22 Comparison of lip recognition accuracy of different models

Based on the above analysis, although the average recognition rate of LSTM is higher than that of DenseNet, which can be seen in Table 4, this paper chooses to optimize the DenseNet model for several reasons. Firstly, DenseNet is more stable in the later stages of training. Second, DenseNet exhibits a more controllable recognition rate trend relative to LSTM, which helps to improve the predictability of the model. In addition, DenseNet demonstrates strong learning and generalization abilities when dealing with complex datasets. Based on the relevant performance of DenseNet, this study decided to focus on further optimization of the DenseNet model.

Table 4 Initial model results

	Lip-reading Recognition Rate
LSTM	0.354
DenseNet	0.298

4.3 Optimization Approaches

This section will focus on the optimizations of the DenseNet model. In this section, an optimized novel coupled 3D experimental model, 3D_DenseNet, is proposed and optimised in the following aspects, Model tuning hyperparameter optimization and model Optimization. The results of the model simulation experiments will be shown in the third subsection of this section, with some of the methods referenced from the relevant literature ^[10] ^[16].

4.3.1 Model tuning hyperparameter optimization

Because of the similarity between the LRW-1000 and LRW datasets, and the good convergence of LipNet on the LRW public dataset, and the high similarity between LRW and LRW-1000, this paper decides to use LipNet for determining the parameter settings for the training of the model for the simulation experiments.

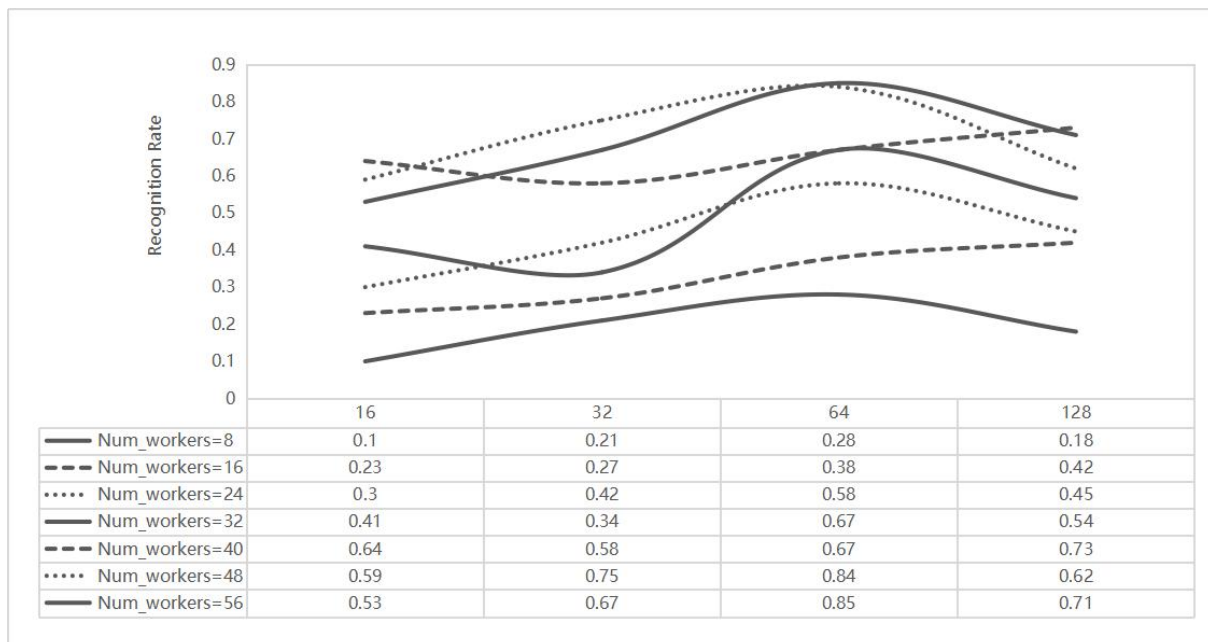
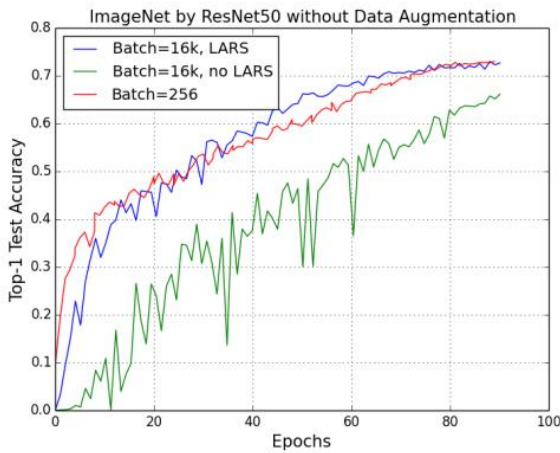


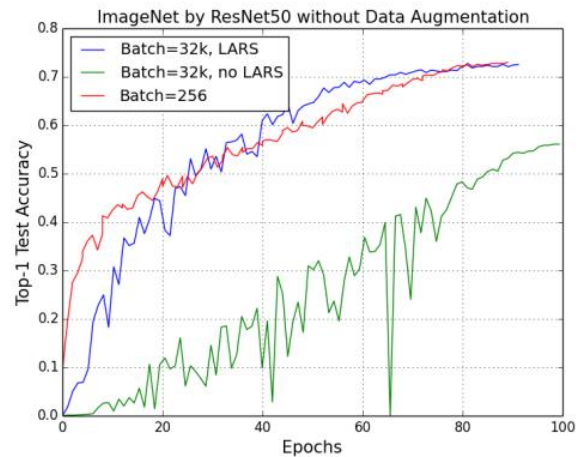
Figure 23 Plot of variation in lip recognition rate for different number of RNN network nodes with different number of threads

The first step is to carry out the setting of the number of nodes of the RNN network. The number of RNN nodes represents the representation capability of the model, i.e., capturing complex patterns and structures in the data. This may affect the features of the data learnt by the model. In general, the increase in the number of nodes can improve the model learning rate. However, at the same time an increase in the number of nodes increases the complexity and computational load of the model and increases the memory requirements, which may require more data for training to avoid overload. Therefore, the experimental setting of this parameter is performed first.

Controlling the irrelevant variable epoch=10, the experiments were conducted by setting the number of RNN network nodes to 16, 32, 64, 128 for the number of iterations epoch=10. Fig. 13 shows the variation of accuracy for different number of RNN network nodes for different number of threads 8, 16, 24, 32, 40, 48, 56 used by the data loader. It can be seen that when the number of nodes is increased from 16 to 32 and from 32 to 64, the overall accuracy rate under all threads becomes an upward trend, whereas when increasing from 64 to 128, most of the recognition rate decreases and it can be seen that, the number of nodes of the RNN network is not higher the better and hence the final choice of the number of nodes is 64. The concrete results are shown in Figure 23 above.



(a) Batch Size = 16k



(b) Batch Size = 32k

Figure 24 Correspondence between epoch and recognition rate for different batch_size. The base learning rate for batch 256 is 0.2 with multiple strategies (power = 2).^{[12] [35]}.

Regarding the epoch parameter setting, due to the limitation of the equipment and the time consuming, this paper adopts the data results obtained from the similar experiments done

by You et al. in the paper "100-epoch ImageNet training with alexnet in 24 minutes" ^[25], and the results are shown below. The results of the experiment are shown in the figure below, and it can be found that under different batch_size conditions, the learning recognition rate tends to be stable when the epoch is close to 100. So, the epoch data of this experiment is set to be higher than 100.

The final settings are shown in the following figure, including batch_size, workers, epoques, paddings, interval, dataroot, gpus, etc.

```
parser = argparse.ArgumentParser('DenseNet 3D')
parser.add_argument('--seed', type=int, default=1)
parser.add_argument('--batch_size', type=int, default=64)
parser.add_argument('--num_workers', type=int, default=24)
parser.add_argument('--epoch', type=int, default=100)
parser.add_argument('--s_epoch', type=int, default=0)
parser.add_argument('--padding', type=int, default=60)
parser.add_argument('--interval', type=int, default=10)

parser.add_argument('--lr', type=float, default=1e-4)
parser.add_argument('--dp', type=float, default=0.2)

parser.add_argument('--gpus', type=str, default='0')
parser.add_argument('--opt', type=str, default='adam')
parser.add_argument('--dataset', type=str, default='LRW-1000')
parser.add_argument('--data_root', type=str, default=r"E:\LRW1000_Public_packed\images")
parser.add_argument('--index_root', type=str, default=r"E:\LRW1000_Public_packed\info")
parser.add_argument('--model_path', type=str, default='')
parser.add_argument('--save_path', type=str, default='weights')

parser.add_argument('--load_weights', action='store_true', default=False)
parser.add_argument('--every_frame', action='store_true', default=False)
parser.add_argument('--usecuda', action='store_false', default=True)
parser.add_argument('--no_train', action='store_true', default=False)
parser.add_argument('--no_val', action='store_true', default=False)
```

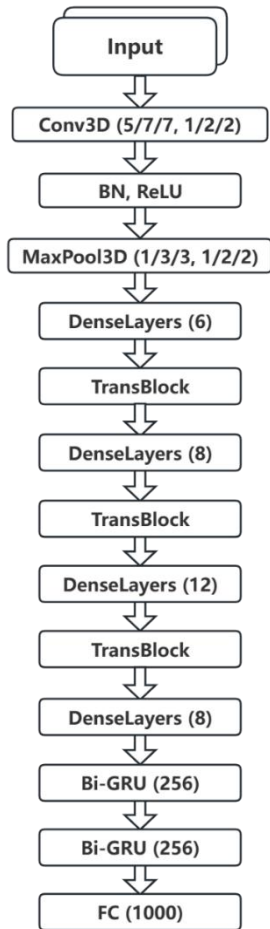
Figure 25 Correspondence between epoch and recognition rate for different batch_size. The base learning rate for batch 256 is 0.2 with multiple strategies (power = 2).^[26]

4.3.2 Model optimization strategy

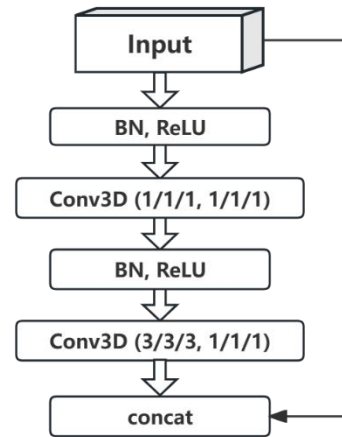
Perhaps the three spatio-temporal layers cannot be learnt due to the strong wildness and variability of the LRW-1000 dataset. The optimization strategy adopted in this paper is to optimize the 2D DenseNet model into a 3D DenseNet model. Although the LipNet network model could not converge on LRW-1000, it achieved good results on the publicly available LRW dataset. Furthermore, since there is a high degree of similarity between the LRW dataset

and the LRW-1000 dataset, this paper attempts to combine these two models to improve the recognition rate. The optimized model is referred to as 3D_DenseNet in the following. Figure 26 shows the process architecture diagram of the 3D_DenseNet model.

(a) The proposed D3D model



(b) The structure of each DenseLayer



(c) The structure of TransBlock

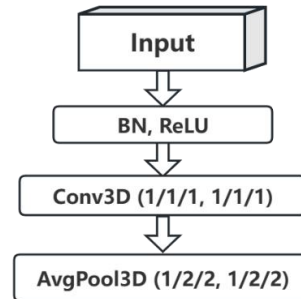


Figure 26 Optimized DenseNet processing framework for 3D situations

Next, it will be shown how this paper implements the whole process of changing the 2D DenseNet structure to a 3D architecture, adjusting the network parameters according to the data dimensions, and how to fuse the 3D DenseNet model with the LipNet model.

(1) Modify the DenseNet network structure to convert to 3D: First replace all 2D convolutional layers with nn. Conv3d to accommodate 3D data input. The processing unit for the time dimension is then added to the original DenseNet model, and the experiments in this paper make use of operations such as 3D convolution and 3D pooling. Implement in Figure 27.

```
self.conv1 = nn.Conv3d(in_channels=3, out_channels=64, kernel_size=(3, 3, 3), stride=(1, 1, 1), padding=(1, 1, 1))
self.pool1 = nn.MaxPool3d(kernel_size=(2, 2, 2), stride=(2, 2, 2))
```

Figure 27 2D DenseNet convert to 3D DenseNet

(2) Adjustment of network parameters: According to the dimension of the input data and the change of the network structure, adjust the network parameters, including the size of the convolution kernel, the number of convolution kernels, the step size, padding and other parameters. The specific parameter adjustment is shown in the following Figure 28.

```
class DenseNet3D(nn.Module):
    def __init__(self, in_channels, out_channels):
        super(DenseNet3D, self).__init__()
        # Define 3D convolutional layers with parameters tuned to the dimensionality of the input data
        self.conv1 = nn.Conv3d(in_channels, out_channels, kernel_size=(3, 3, 3), stride=1, padding=1)

# Specify the number of input channels and the number of output channels
model = DenseNet3D(in_channels=3, out_channels=64)
```

Figure 28 Optimized network parameters

(3) Selection of loss function and optimizer: For the 3D DenseNet model, the cross-entropy loss function is chosen for the optimisation experiments in this paper. The cross-entropy loss function was first defined as the loss function for the multi-classification task using `nn.CrossEntropyLoss()`. Then the Adam optimiser was chosen as the optimiser, defined through `optim`. During training, the loss values are backpropagated and parameter updates are performed through the optimizer.

```
# Define the loss function
criterion = nn.CrossEntropyLoss()
# Defining the Optimiser
optimizer = optim.Adam(model.parameters(), lr=0.001) # 使用Adam优化器

for epoch in range(num_epochs):
    running_loss = 0.0
    for i, data in enumerate(train_loader, 0):
        inputs, labels = data['image'], data['label']
        optimizer.zero_grad()
        outputs = model(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item()
    if i % 100 == 99:
        print('[%d, %5d] loss: %.3f' %
              (epoch + 1, i + 1, running_loss / 100))
        running_loss = 0.0
```

Figure 29 Loss function and optimizer selection

(4) Fusion with LipNet models: In this paper, a feature-level model fusion approach is used to fuse the 3D DenseNet model with the LipNet model. The outputs of the two models are combined through the fusion layer and then fed into the fully connected layer for final classification to predict the outcome. Specifically, the respective feature representations are first extracted from each model. For the 3D DenseNet model, this study extracts video sequence-based feature representations, while for the LipNet model, lip-speak-based feature representations are extracted. These features are then fused so that the features extracted by the two models are joined to form a larger feature vector. Finally, the weight file of the fused model is adjusted based on experiment-by-experiment. The weight file includes parameters such as weight coefficients and biases for each layer of the entire optimisation model. In this paper, the splicing operation before the global pooling layer is used to fuse the features of the two models. Which fuses the features before the global pooling layer as well as after the Bi-LSTM layer. Finally, a fully connected layer was used to map the fused features to the output categories.

Before using this model, the research presupposes ensuring that the output shapes of the two models are consistent so that feature fusion can be performed correctly. During training, the research implemented the fusion strategy by dynamically adjusting the weights of the two models by the outputs of both models in the loss function. The code period implementation in the following Figure 30.

```
class FusionModel(nn.Module):
    def __init__(self, dense_net, lip_net, num_classes):
        super(FusionModel, self).__init__()
        self.dense_net = dense_net
        self.lip_net = lip_net
        self.fc = nn.Linear(3 * num_classes, num_classes) # Add the feature length of the output of the two models
        1 usage (1 dynamic)
    def forward(self, image, lip_features):
        dense_features = self.dense_net(image)
        lip_output = self.lip_net(lip_features)
        # Fusion layer: Concatenates the output of two models
        combined_features = torch.cat((dense_features, lip_output), dim=1)
        # The fully connected layer is classified
        output = self.fc(combined_features)
        return output

dense_net = DenseNet3D(in_channels=3, out_channels=64)
lip_net = LipNet(input_size, hidden_size, num_layers, num_classes)
fusion_model = FusionModel(dense_net, lip_net, num_classes)
```

Figure 30 Loss function and optimizer selection

4.3.3 Optimized model training rate

As can be seen from the figure below, the recognition rate of the validation set basically fluctuates up and down at 46.73446% when the epochs are greater than 100, while the recognition rate of the test set shows a gradual increase and grows in a logarithmic trend. The recognition rate of the test set fluctuates around 20% with no more than 10% up and down. As the model learns and understands the features of the dataset more deeply in the later stages of training, this leads to the stabilization and improvement of the recognition rate in both the validation and test sets.

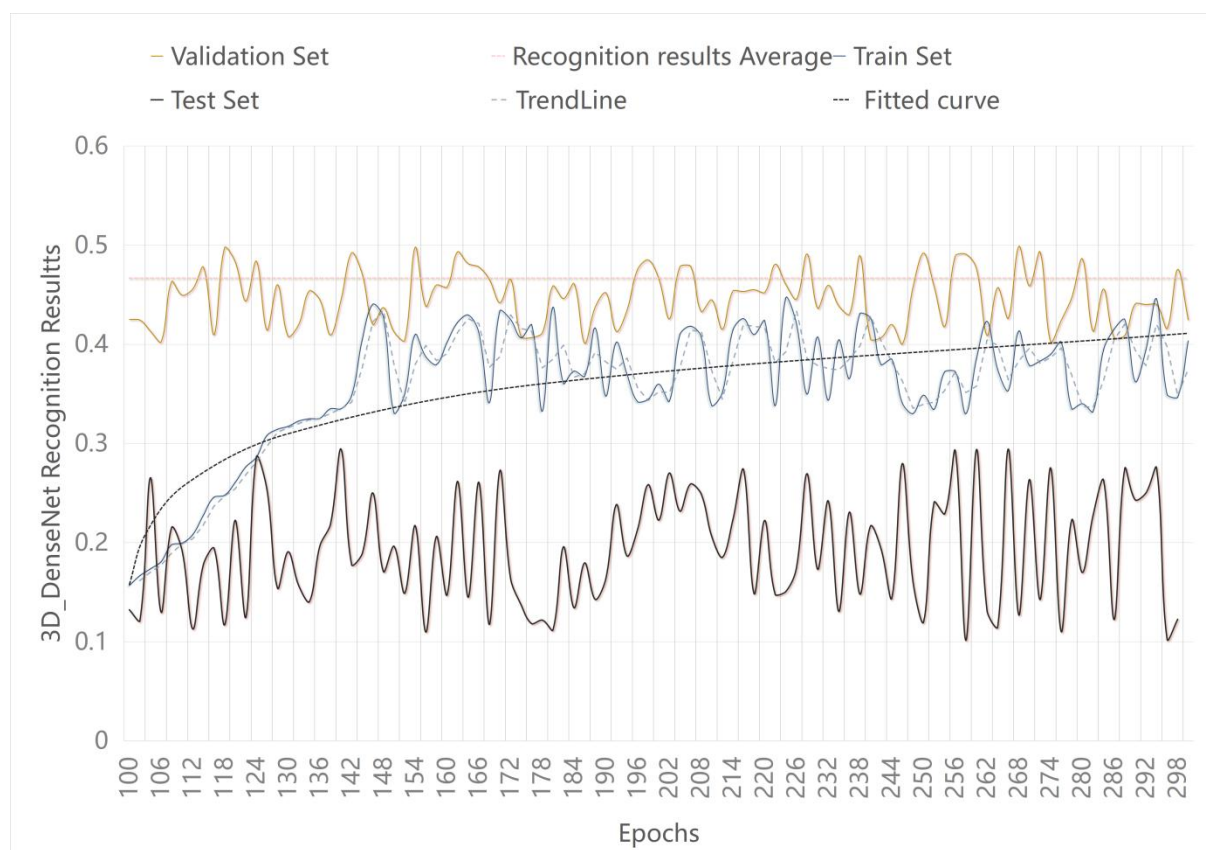


Figure 31 3D_DenseNet model data training test validation results

It is clearly seen that the recognition rate of the optimized 3D DenseNet model has increased by about 20% or so compared to the previous one, indicating that the model has achieved higher learning and generalization capabilities. Although the recognition rate is still not high, compared with the highest recognition rate of 52.7% for the current technique on LRW-1000, the model has basically achieved the optimization goal. The 3D DenseNet model has a better capture and understanding of the dataset features, and the model can cope with the challenges of the LRW-1000 dataset more effectively and achieve better recognition results.

5 CONCLUSIONS

5.1 Review of Aims

Lip recognition is an important and popular research topic in the field of deep learning, so the research on lip recognition techniques is of great significance. In addition, although satisfactory results have been achieved in model training recognition rates for lip-speaking word-level and sentence-level datasets in single and laboratory environments, current research has not yielded good experimental results for datasets in complex environments and real-world situations.

In view of this, our study aims to compare the recognition accuracy of different lip recognition models on datasets in complex environments, adjust the hyper-parameters of the data, etc., and propose an optimized lip recognition model to improve the recognition rate on datasets close to real-world environments. Models such as LSTM, LipNet, and DenseNet are the classical algorithms proposed for lip recognition in recent years. In this paper, we firstly select a highly variable and wild lip recognition dataset LRW-1000 to simulate the lip recognition in the real situation and carry out simulation experiments to compare the recognition rates of the current LSTM, LipNet, DenseNet and other models under this dataset. It is found that the recognition rate of several models on the LRW-1000 dataset is not high. Therefore, an optimized model for 3D_DenseNet is proposed in this study. In this paper, a feature-level model fusion approach is used to fuse the 3D DenseNet model with the LipNet model. The outputs of the two models are combined through the fusion layer and then fed into the fully connected layer for final classification to predict the results. Specifically, the respective feature representations are first extracted from each model. For the 3D DenseNet model, this study extracts video sequence-based feature representations, while for the LipNet model, lip-speak-based feature representations are extracted. These features are then fused so that the features extracted by the two models are joined to form a larger feature vector. Finally, the weight files of the fused model are adjusted based on experiment-by-experiment. The results show that the results show that the optimization model is more adaptable in complex environments and has better generalizations capabilities. Overall, the objectives of this project have been achieved.

5.2 Future Work

The optimization model 3D_DenseNet proposed in this paper provides an effective research method in the field of lip recognition and contributes to the development of lip recognition technology. Due to the complexity of the LRW-1000 dataset, this paper performs processes such as cleaning and augmentation of the data, aiming to reduce the complexity of the dataset to facilitate model learning and processing. However, this process may have reduced the realism of the data. In order to better simulate real-world situations, lip-speaking datasets in more complex environments need to be actively explored and adequately cleaned and labelled to ensure the quality and authenticity of the data.

Meanwhile, for the problem that the recognition rate of the current model on this dataset is not high, it becomes an important task to further improve the performance of the model. In this process, the introduction of new deep learning techniques, such as Attention Mechanism and Transformer, can be considered to enhance the performance of the model in complex environments.

In addition, in the future, it is hoped that the optimized model will be applied to real-world scenarios and its performance will be verified and tested to further validate its feasibility and effectiveness in real-world applications.

Although this paper has made significant progress in the field of lip recognition, there is still room for improvement. Through further research work and practical exploration, the model is expected to make greater breakthroughs in the field of lip recognition. The implementation of these future works will help to improve the recognition rate and efficiency of the model and provide important technical support and guarantee for the realization of smarter and more convenient human-computer interaction.

References

- [1] Afouras, T., Chung, J. S., Zisserman, A. Deep lip reading: a comparison of models and an online application[C]. In: Conference of the International Speech Communication Association 2018. pp. 3244-3248.
- [2] Assael, Y. M., Shillingford, B., and Whiteson, S., et al. LipNet: End-to-end sentence-level lipreading[J]. arXiv preprint arXiv:1611.01599, 2016.
- [3] Chan, Z. M., Lau, C. Y., Thang, K. F. Visual Speech Recognition of Lips Images Using Convolutional Neural Network in VGG-M Model[J]. J. Inf. Hiding Multim. Signal Process., 2020. 11(3): 116-125.
- [4] Chung, J. S., Zisserman, A. Out of time: automated lip sync in the wild[C]. Asian Conference on Computer Vision. Berlin: Springer, 2016. pp. 251-263.
- [5] Chung, J., S., Senior, A., and Vinyals, O., et al. Lip reading sentences in the wild[C]. 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2017. pp. 3444-3453.
- [6] Chung, J. S., Zisserman, A. Lip Reading in Profile [C]. British Machine Vision Conference, 2017.
- [7] Chung, J. S., Zisserman, A. Learning to lip read words by watching videos[J]. Computer Vision and Image Understanding, 2018, 173. pp. 76-85.
- [8] Chung, J., S., and Zisserman, A. Lip reading in the wild[C]. Asian Conference on Computer Vision. Springer, Cham, 2016. pp. 87-103.
- [9] Cox, S. J., Harvey, R. W., Lan, Y., et al. The challenge of multispeaker lip-reading[C]. AVSP. 2008. pp. 179-184.
- [10] Dai, Y. et al. Improving audio-visual speech recognition by lip-subword correlation based visual pre-training and cross-modal fusion encoder[C]. In: Proceedings of 2023 IEEE International Conference on Multimedia and Expo (ICME). IEEE, 2023. 2627-2632.
- [11] Fu, Y., Yan, S., Huang, T. S. Classification and feature extraction by simplexization[J]. IEEE Transactions on Information Forensics and Security, 2008. pp. 91-100.
- [12] Goyal, P.; Dollar, P.; Girshick, R.; Noordhuis, P.; Wesolowski, L.; Kyrola, A.; Tulloch, A.; Jia, Y.; and He, K. 2017. Accurate, large minibatch sgd: Training imagenet in 1 hour[J]. arXiv preprint arXiv:1706.02677.
- [13] Graves, A., Fernández, S., Gomez, F., et al. Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks[C]. Proceedings of the 23rd International Conference on Machine Learning. ACM, 2006. pp. 369-376.
- [14] Hochreiter, S., Schmidhuber, J. Long short-term memory[J]. Neural computation, 1997, 9(8).

- pp. 1735-1780.
- [15] Hu, D., Li, X. Temporal multimodal learning in audiovisual speech recognition[C]. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016. pp. 3574-3582.
 - [16] Huang H, Song C, Ting J, et al. A novel machine lip reading model[J]. Procedia Computer Science, 2022, 199: 1432-1437.
 - [17] Jeon S, Kim M S. End-to-end lip-reading open cloud-based speech architecture[J]. Sensors, 2022, 22(8): 2938.
 - [18] Kazemi V, Sullivan J. One millisecond face alignment with an ensemble of regression trees [C]. In Proceedings of the IEEE conference on computer vision and pattern recognition, 2014: 1867-1874
 - [19] Kumar L A, Renuka D K, Rose S L, et al. Deep learning based assistive technology on audio visual speech recognition for hearing impaired[J]. International Journal of Cognitive Computing in Engineering, 2022, 3: 24-30.
 - [20] Lee, B., Hasegawa-Johnson, M., Goudeseune, C., et al. AVICAR: Audio-visual speech corpus in a car environment[C]. Eighth International Conference on Spoken Language Processing. 2004.
 - [21] Liu L, Peng Y, Liu M, et al. Sensor-based human activity recognition system with a multilayered model using time series shapelets[J]. Knowledge-Based Systems, 2015, 90: 138-152.
 - [22] Lu Y, Li H. Automatic lip-reading system based on deep convolutional neural network and attention-based long short-term memory[J]. Applied Sciences, 2019, 9(8): 1599.
 - [23] Lu Y, Tian H, Cheng J, et al. Decoding lip language using triboelectric sensors with deep learning[J]. Nature communications, 2022, 13(1): 1401.
 - [24] McGurk, H., MacDonald, J. Hearing lips and seeing voices[J]. Nature, 1976, 264(5588). pp. 746.
 - [25] Noda, K., Yamaguchi, Y., and Nakadai, K., et al. Lipreading using convolutional neural network[C]. Fifteenth Annual Conference of the International Speech Communication Association. 2014.
 - [26] Papandreou, G., Katsamanis, A., Pitsikalis, V., et al. Adaptive multimodal fusion by uncertainty compensation with application to audiovisual speech recognition[J]. IEEE Transactions on Audio, Speech, and Language Processing, 2009. pp. 423-435.
 - [27] Patterson, E, K., Gurbuz, S., Tufekci, Z., et al. CUAVE: A new audio-visual database for multimodal human-computer interface research[C]. 2002 IEEE International Conference on Acoustics, Speech, and Signal Processing. IEEE, 2002, 2. pp. II-2017-II-2020.

-
- [28] Saitoh, T., Zhou, Z., Zhao, G., et al. Concatenated frame image based cnn for visual speech recognition[C]. Asian Conference on Computer Vision. Springer, Berlin, 2016. pp. 277-289.
- [29] Stafylakis, T., Tzimiropoulos, G. Combining residual networks with LSTMs for lipreading[J]. In: Interspeech, August 2017, Stockholm, Sweden 2017, 20-24.
- [30] Shillingford, B., Assael, Y., Hoffman, M. W., et al. Large-scale visual speech recognition[J]. arXiv preprint arXiv:1807.05162, 2018.
- [31] Vaswani, A., Shazeer, N., Parmar, N., et al. Attention is all you need[C]. Advances in Neural Information Processing Systems. 2017. pp. 5998-6008.
- [32] Wand, M., Koutnik, J., and Schmidhuber, J. Lipreading with long short-term memory[C]. 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2016. pp. 6115-6119.
- [33] Wu, K. C. Lip recognition based on deep learning[D]. MSc. Dissertation: Southeast University, 2020. (In Chinese).
(吴开存. 基于深度学习的唇语识别[D]. 东南大学, 2020.)
- [34] Yang S, Zhang Y, Feng D, et al. LRW-1000: A Naturally-Distributed Large-Scale Benchmark for Lip Reading in the Wild[C]. 2019 14th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2019). IEEE, 2019. pp. 1-8.
- [35] You Y, Zhang Z, Hsieh C, et al. 100-epoch imagenet training with alexnet in 24 minutes[J]. arXiv preprint arXiv:1709.05011, 2017, 8.
- [36] Zhang, Y. Multidimensional time series classification and its application in lip recognition[D]. MSc. Dissertation, Beijing: Beijing Jiaotong University, 2012. Doi: 10.26944/d.cnki.gbfju.2021.002043. (In Chinese).
(张月. 多维时间序列分类方法及其在唇语识别上的应用研究[D]. 北京交通大学, 2021.)

Acknowledgements

At this point, I am about to end my undergraduate life, looking back on the four years of college, my heart is full of feelings and gratitude.

Firstly, I express my gratitude to all the teachers who have provided me with assistance and guidance. Especially, I would like to thank Mr. Wang Zhihai for his care and support when I changed my major in my freshman year, and the teachers of my major courses for their guidance and direction in my study of computer science. I would also like to express my heartfelt thanks to Mr. Liu Jie for providing me with program opportunities and daily encouragement.

Secondly, I would like to thank all the friends I have made on campus. I would like to thank the group members who struggled and worked hard with me, and I would like to thank Yijun for inviting me to form a team together for the first time in the summer of my sophomore year and for always being the group's solid backing. I can say that it has helped me a lot in my professional skills. She has been a tremendous help in improving my professional skills. In addition, I would like to thank all the seniors who have helped me with their role model power and selfless sharing and guidance. I am grateful for every joy and warmth in my university life. Special thanks to Xuehan, whose help after I switched to the computer science major enabled me to adapt to the new environment quickly and gave me endless care and support in the subsequent friendship.

Thanks to Shiqi, we supported each other through the four years of growing up together, and this friendship will be treasured in my heart forever. Also, thank you to all my friends who have given me joy.

I would like to give special thanks to my parents, who have always supported and trusted me unconditionally, giving me advice and encouragement. Their companionship and warmth are the inexhaustible motivation for me to move forward.

Finally, I would like to thank myself. I would like to thank myself for the efforts I have made in my studies day after day, year after year, and for every choice I have made. I hope that when I look back on my four years of college in the future, I will have no regrets in my heart; I also hope that I will always keep my beginnings and happiness. May my friends and I continue to work hard on the road of life in the future..

Appendixes

Appendix A: 3D_DenseNet model

```

7  class NLLSequenceLoss(nn.Module):
8      """
9      Custom loss function.
10     Returns a loss that is the sum of all losses at each time step.
11     """
12     def __init__(self):
13         super(NLLSequenceLoss, self).__init__()
14         self.criterion = nn.NLLLoss(reduction='none')
15
16     def forward(self, input, length, target, every_frame=False):
17         loss = []
18         transposed = input.transpose(0, 1).contiguous()
19         for i in range(transposed.size(0)):
20             loss.append(self.criterion(transposed[i, :], target).unsqueeze(1))
21         loss = torch.cat(loss, dim=1)
22         mask = torch.zeros(loss.size(0), loss.size(1)).float().cuda()
23
24         for i in range(length.size(0)):
25             L = min(mask.size(1), length[i])
26             if every_frame:
27                 mask[i, :L] = 1.0
28             else:
29                 mask[i, L-1] = 1.0
30         loss = (loss * mask).sum() / mask.sum()
31         return loss

```

1 usage

```

35 def _validate(modelOutput, length, labels, every_frame=False):
36     labels = labels.cpu()
37     averageEnergies = torch.zeros((modelOutput.size(0), modelOutput.size(-1)))
38     for i in range(modelOutput.size(0)):
39         if every_frame:
40             averageEnergies[i] = torch.mean(modelOutput[i, :length[i]], 0)
41         else:
42             averageEnergies[i] = modelOutput[i, length[i] - 1]
43
44     _, maxindices = torch.max(averageEnergies, 1)
45     count = torch.sum(maxindices == labels)
46     return count

```

2 usages

```

49 class LipReading(torch.nn.Module):
50     def __init__(self, growth_rate=32, num_init_features=64, bn_size=4, drop_rate=0.2, num_classes=1000):
51         super(LipReading, self).__init__()
52         #block_config = (6, 12, 24, 16)
53         block_config = (4, 8, 12, 8)

```

```

54
55 self.features = nn.Sequential(OrderedDict([
56     ('conv0', nn.Conv3d(in_channels=3, num_init_features, kernel_size=(5, 7, 7), stride=(1, 2, 2), padding=(2, 3, 3), bias=False)),
57     ('norm0', nn.BatchNorm3d(num_init_features)),
58     ('relu0', nn.ReLU(inplace=True)),
59     ('pool0', nn.MaxPool3d(kernel_size=(1, 3, 3), stride=(1, 2, 2), padding=(0, 1, 1))),
60 ]))
61 num_features = num_init_features
62 for i, num_layers in enumerate(block_config):
63
64     block = _DenseBlock(num_layers=num_layers, num_input_features=num_features,
65                         bn_size=bn_size, growth_rate=growth_rate, drop_rate=drop_rate)
66     self.features.add_module('denseblock%d' % (i + 1), block)
67
68     num_features = num_features + num_layers * growth_rate
69     if i != len(block_config) - 1:
70         trans = _Transition(num_input_features=num_features, num_output_features=num_features // 2)
71         self.features.add_module('transition%d' % (i + 1), trans)
72         num_features = num_features // 2
73
74     # Final batch norm
75
76     self.features.add_module(name='norm5', nn.BatchNorm3d(num_features))
77     self.gru1 = nn.GRU(536*3*3, hidden_size=256, bidirectional=True, batch_first=True)
78     self.gru2 = nn.GRU(input_size=512, hidden_size=256, bidirectional=True, batch_first=True)
79
80     self.fc = nn.Sequential(
81         nn.Dropout(0.5),
82         nn.Linear(in_features=512, num_classes)
83     )
84     self.loss = NLLSequenceLoss
85     self._initialize_weights()
86
87     4 usages (4 dynamic)
88     def validator_function(self):
89         return _validate
90
91     def _initialize_weights(self):
92         for m in self.modules():
93             if isinstance(m, nn.Conv3d):
94                 n = m.kernel_size[0] * m.kernel_size[1] * m.kernel_size[2] * m.out_channels
95                 m.weight.data.normal_(mean=0, math.sqrt(2. / n))
96                 if m.bias is not None:
97                     m.bias.data.zero_()
98             elif isinstance(m, nn.Conv2d):
99                 n = m.kernel_size[0] * m.kernel_size[1] * m.out_channels
100                m.weight.data.normal_(mean=0, math.sqrt(2. / n))
101                if m.bias is not None:
102                    m.bias.data.zero_()
103            elif isinstance(m, nn.Conv1d):
104                n = m.kernel_size[0] * m.out_channels
105                m.weight.data.normal_(mean=0, math.sqrt(2. / n))
106                if m.bias is not None:
107                    m.bias.data.zero_()
108            elif isinstance(m, nn.BatchNorm3d):
109                m.weight.data.fill_(1)
110                m.bias.data.zero_()

```

```
114         elif isinstance(m, nn.BatchNorm2d):
115             m.weight.data.fill_(1)
116             m.bias.data.zero_()
117
118         elif isinstance(m, nn.BatchNorm1d):
119             m.weight.data.fill_(1)
120             m.bias.data.zero_()
121
122     def forward(self, x):
123         self.gru1.flatten_parameters()
124         self.gru2.flatten_parameters()
125         f2 = self.features(x)
126         f2 = f2.permute(0, 2, 1, 3, 4).contiguous()
127         B, T, C, H, W = f2.size()
128         f2 = f2.view(B, T, -1)
129         f2, _ = self.gru1(f2)
130         f2, _ = self.gru2(f2)
131         f2 = self.fc(f2).log_softmax(-1)
132         return f2
```